

The Star–Planet Composition Connection

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Abstract

The mantra “know thy star, know thy planet” has proven to be very important for many aspects of exoplanet science. Here I review how stellar abundances inform our understanding of planet composition and, thus, formation and evolution. In particular, I discuss how:

- The strongest star–planet connection is still the giant planet–metallicity correlation, the strength of which may indicate a break point between the formation of planets versus brown dwarfs.
- We do not have very good constraints on the lower metallicity limit for planet formation, although new statistics from TESS are helping, and it appears that, at low $[\text{Fe}/\text{H}]$, α elements can substitute for iron as seeds for planet formation.
- The depletion of refractory versus volatile elements in stellar photospheres (particularly the Sun) was initially suggested as a sign of small planet formation but is challenging to interpret, and small differences in binary star compositions can be attributed mostly to processes other than planet formation.
- We can and should go beyond comparisons of the carbon-to-oxygen ratio in giant planets and their host stars, incorporating other volatile and refractory species to better constrain planet formation pathways.
- There appears to be a positive correlation between small planet bulk density and host star metallicity, but exactly how closely small planet refractory compositions match those of their host stars—and their true diversity—is still uncertain.

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1. INTRODUCTION

To the best of our knowledge, planets generally form in orbit around stars in disks composed of the gas, dust, and ice left over from star formation, which itself holds a record of billions of years of galactic and stellar evolution. Until approximately 30 years ago, we had only one example of a successful iteration of this process around a Sun-like star, although this example provides the most detailed sources for learning about the outcome of planet formation in Solar System planets, moons, comets, asteroids, and so forth. Now, with more than 5,600 confirmed planets around other stars (see <https://exoplanetarchive.ipac.caltech.edu>), many in multiplanet systems, it is clear that (a) planet formation is common in the Galaxy and can occur around a range of stellar types, (b) the architectures of exoplanet systems can be very different from that of the Solar System, and (c) there is a wider range of bulk densities (a rough proxy for composition) in exoplanets than that found in Solar System planets. Much of the research in the field of exoplanets is related to understanding the factors and processes driving these three points and how “normal” the Solar System is in terms of how it formed and the resulting planet compositions, including whether or not they present conditions suitable for the emergence of life.

Host star properties like mass (see Mulders 2018 and references therein), activity and age (e.g., Lecavelier des Etangs et al. 2004, Linsky 2014, Christensen-Dalsgaard & Aguirre 2018), and multiplicity (see Howell et al. 2022 and references therein) play significant roles in the characteristics of the exoplanet population we observe today. Obtaining accurate and precise host star properties is also critical for extracting accurate and precise planet properties—deriving fundamental parameters like planet mass (M_p) and planet radius (R_p) from observations requires knowledge of the host star—and some specific host star properties directly influence the efficacy of exoplanet detection (meaning that, e.g., we are biased in our exoplanet sample toward inactive and single or wide binary stars).

This review focuses on how host star composition (which, of course, is not independent of other properties like stellar mass and age) is related to planet properties. Understanding the origins of different stellar compositions is a fascinating and dynamic field unto itself, but suffice it to say that stars across the Galaxy display a range of different abundances related to their time and location of formation. In a simplistic sense, considering all populations of stars in the Milky Way,

the abundances of elements heavier than H and He (i.e., metals) increase with time, so that the oldest stars are the most metal-poor and the youngest are the most metal-rich. In actuality, the trends between metallicity and age are more complex and related to other variables like $[\alpha/\text{Fe}]$ (discussed in Section 2.2) and stellar orbital properties (Feuillet et al. 2018, Xiang & Rix 2022). The relative proportions of metals are dictated by the astrophysical processes producing them. From here onward, I use the term metallicity to indicate the iron abundance, $[\text{Fe}/\text{H}]$, of a star. This bracket notation indicates the log abundance relative to the Sun, that is, $[A/B] \equiv \log_{10}(N_A/N_B)_* - \log_{10}(N_A/N_B)_\odot$, where N_A and N_B are the number densities of elements A and B. The increase in Fe over time is due to its production primarily in type Ia supernovae (SNe), wherein white dwarfs that used to be low-mass stars accrete enough mass from a companion star to trigger C fusion and runaway thermonuclear explosion. Other elements like Mg, Si, and Ca (called α elements because they form via the α -capture process) are produced primarily in massive stars and dispersed earlier by type II SNe, so the $[\alpha/\text{Fe}]$ ratio is an important tracer of different stellar populations. The elements O and S are also often included with α elements, and both generally decrease relative to Fe with time; the source of C is less certain (type II SNe, novae, massive star winds, low- to intermediate-mass stars), but it changes slightly more than O with age such that C/O has a small positive slope with age (Nissen et al. 2014, Bedell et al. 2018). For a much more in-depth discussion of the process of Galactic chemical evolution (GCE; dynamics is also likely important, leading to the term chemodynamical evolution), see McWilliam (1997), Frebel & Norris (2013), Yates et al. (2013), Feuillet et al. (2018), Ness et al. (2019), Matteucci (2021), and Xiang & Rix (2022).

This review approaches the connection between star and planet compositions from two related perspectives. First, how are stellar compositions related to the presence or specific architectures of planetary systems (Section 2)? Second, in what ways do host star compositions inform our interpretation or prediction of planet compositions (Section 4)? Section 3 takes a detour into a few special cases of anomalous abundances—how the formation or presence of planets can change a star’s abundances in a measurable way in comparison to a star without known planets, although of course we now know that most stars host planets (e.g., Kunimoto & Matthews 2020) and it is actually difficult to pinpoint stars that definitely do not host planets. The review concludes with a few future prospects for new or enhanced ways of using stellar abundances to learn about planet compositions (Section 5).

Before diving in, I give a very brief overview of how host star abundances are derived. For excellent reviews of the process of deriving stellar abundances, see Nissen & Gustafsson (2018) and Jofré et al. (2019).

Stellar abundances, representative of the average photosphere, are derived from spectra, which need to be of high enough resolution to separate or measure (most) individual line intensities. Higher resolving power and signal-to-noise (S/N) spectra allow for more in-depth characterization of the star, including measurement of more elements that have weaker lines. Typically, the minimum values for abundance derivation for reaching the precision limits we have today are around $R \gtrsim 20,000$ and $S/N \gtrsim 100$, although they depend on the specific case (e.g., a small number of abundances but across many thousands of stars). There are nuances in both reducing spectral data (e.g., wavelength calibration, merging orders in the case of echelle spectra, cosmic ray correction) and preparing them for the extraction of information (e.g., subtracting telluric absorption from Earth’s atmosphere and normalization, namely blaze function removal). Most abundance research is done with moderate- to high-resolution optical spectra, but other wavelength regions can be more appropriate depending on the object, for example, near-IR (NIR) for M dwarf stars or UV for polluted white dwarf stars.

The basic measurements used to derive abundances from spectra are the strengths of lines from different species. Line strengths can be based on the measurement of equivalent widths (EWs),

by fitting a Gaussian or Voigt profile or simply integrating over the line profile. These are then compared with the curve of growth associated with specific stellar parameters, where for lines that are not saturated there is a linear relationship between the abundance (number of atoms) and the EW. This method is sensitive to how the continuum around the fitted line is chosen and is not ideal for blended or broadened lines that can give overestimated abundances. Alternatively, line strengths can be compared with synthetic profiles (usually from a predetermined synthetic spectrum with given stellar parameters) of varying abundances until the best match is found. An extension of the line-specific synthesis method is comparing a large part of the (or the entire) spectrum with a library of synthetic stellar spectra, or even a library of observed spectra of stars with well-calibrated parameters. However, in practice, synthesis is typically used for special cases of blended or very weak lines, crowded features, and molecular features. The main downside of synthesis analysis is that it is very sensitive to the instrument profile and the wavelength solution, as every point in the observed line profile is fitted; the instrumental profile and broadening due to rotation and microturbulence have to be carefully included in the full synthesis model, whereas EWs do not require this step. Spectral synthesis results can also suffer from systematic bias because of correlations between T_{eff} , $[\text{Fe}/\text{H}]$, and $\log g$ when solving for all three quantities at once, which is less of an issue in the curve-of-growth technique (Torres et al. 2012). Given the growing number of very large data sets of stellar spectra, there is increasing interest in using machine learning approaches to measuring abundances, although these still rely on a robust training set of labels from either a subset of the observed spectra or synthetic spectra (Ness et al. 2015, Ting et al. 2019).

Translating these line measurements into actual abundances requires an atmospheric model. The most widely used models are still calculated under the assumption of 1D local thermodynamic equilibrium (LTE), but the availability of 3D non-LTE models is increasing, and comparisons between the two have shown the necessity of using the latter for at least some lines (e.g., 7771–5 Å O I triplet; for a recent calculation of corrections between 3D non-LTE and 1D LTE, see, e.g., Amarsi et al. 2019) or classes of spectra (e.g., ultra-metal-poor stars). Modeling the atmospheres of cool stars ($\lesssim 4,000$ K) is challenging due to the prevalence of molecular species in the spectra and a lack of accurate line parameters (position and transition probability, the latter a product of the oscillator strength and statistical weight), which require laboratory spectroscopy, theoretical calculations, or astrophysical calibration (combining a calibrated model with an observed line of uncertain oscillator strength). Indeed, even many atomic lines at optical wavelengths have $\geq 10\%$ uncertainty in their transition probabilities. Thus, not surprisingly, which lines one measures can influence the resulting abundances. Lines of a given element spanning a range of excitation potentials and ionization states help probe different parts of the stellar atmosphere, and the goal is to obtain an abundance that is well representative of the average. The most precise stellar abundances [and parameters;¹ see Jofré et al. (2019, suppl. text 1)] are derived from differential analyses, particularly of similar stars. This approach reduces systematic uncertainties arising from the choice of continuum as well as from line selection, as a line-by-line comparison effectively cancels out uncertainties from non-LTE, blends, poorly constrained line parameters, and so on.

Most of the analyses discussed in this article focus on FGK dwarf stars, leaving out cooler and lower-mass M dwarfs. Measuring accurate and precise abundances (and stellar parameters) for

¹ Here I refer to effective temperature T_{eff} , surface gravity g (usually given as $\log g$), and microturbulent velocity ξ , in addition to $[\text{Fe}/\text{H}]$. Deriving these parameters can be and often is also done from the spectra by measuring many Fe lines and by (a) minimizing the dependence of $[\text{Fe}/\text{H}]$ on the excitation potential χ of the lines (which sets T_{eff}), (b) requiring $[\text{Fe}/\text{H}]$ derived from Fe I and Fe II lines to agree (which sets $\log g$), and (c) minimizing the dependence of $[\text{Fe}/\text{H}]$ on reduced EWs, $\log(\text{EW}/\lambda)$ (which sets microturbulence). Other methods for deriving T_{eff} and $\log g$ exist that are independent of the spectra and thus can provide an extra check.

M dwarfs presents extra challenges because of their generally fainter magnitudes in the optical range, where many high-resolution spectrographs operate, and optical spectra overlain with molecular features that blend in with atomic features and make it hard to determine the continuum. Progress has been made by benchmarking abundances of M dwarfs in binary systems to those of their more massive companions (e.g., Maldonado et al. 2020, Woolf & Wallerstein 2020) and/or by using NIR spectra which have more flux and fewer strong molecular blends (e.g., Veyette et al. 2017, Ishikawa et al. 2020, Souto et al. 2022); some of these studies have even derived abundances beyond [Fe/H]. And yet, abundance measurements are still lacking for many M dwarf exoplanet host stars. This is an active area of growth in stellar astrophysics and exoplanet science and will likely change rapidly in the coming years (see Section 5).

The take-home message from this section is that there are many choices that go into deriving stellar abundances that can affect their accuracy and precision. The effects that different methodological choices can have on the resulting abundances are discussed in greater detail by Smiljanic et al. (2014) and Hinkel et al. (2016). When studying the relationships between exoplanets and their host stars, it is ideal to use stellar abundances derived from the same source, using the same methodology, which helps at least in a relative comparison sense. It is heartening that multiple large all-sky spectroscopic surveys are commencing or on the horizon, as we may soon have uniform abundances across most exoplanet host stars (Section 5).

2. STELLAR COMPOSITION AS AN INDICATOR OF PLANET PRESENCE AND ARCHITECTURE

Despite the nuances described above, stellar compositions are generally easier to measure than those of their orbiting planets. Thus, there is great potential in using different stellar compositions as indicators of the types of planetary systems that can form or have formed around them. This section presents updates to or fairly new trends in stellar composition versus planet properties that inform how we think about planet formation.

2.1. Metallicity (Iron Abundances)

Soon after the first detections of exoplanets orbiting Sun-like stars, what remains the most influential host star–planet composition connection was identified: Short-period gas giant planets tend to orbit more metal-rich (high [Fe/H] abundance) stars (Gonzalez 1997; Santos et al. 2001, 2004; Fischer & Valenti 2005). As discussed in Section 3.2, the community initially debated whether this trend was a cause or effect of planet formation, but eventually it was interpreted as evidence that the dominant formation mechanism for giant planets (at least those fairly close-in around Sun-like stars) is core accretion (Pollack et al. 1996). Population synthesis modeling of core accretion that includes planet migration, disk evolution, and accounting for detection biases provides a good match to the observed systems (e.g., Ida & Lin 2004; Mordasini et al. 2009, 2012), whereas planet formation via the disk/gravitational instability method is relatively insensitive to the disk opacity, which is likely related to the metallicity of the parent star (Boss 2002). This review focuses on stellar composition, but it is important to remember that stellar metallicity and other properties like mass, luminosity, and age are not independent. In particular, the available mass budget for forming planets via core accretion is a function of both the metallicity and mass of the protoplanetary disk (the latter is often assumed to be related to the stellar mass)—even a relatively high metallicity environment may not form giant planets if there is not enough total mass (Johnson et al. 2010, Mordasini et al. 2012, Ghezzi et al. 2018, Manara et al. 2018, Mulders et al. 2021). The following subsections discuss what is known about the [Fe/H] trends in stars hosting different types of exoplanets.

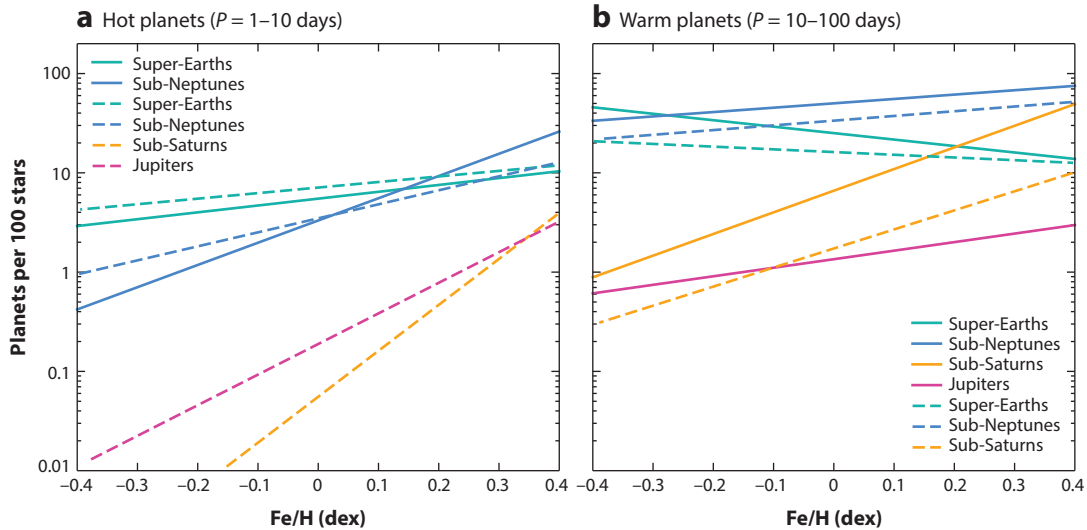


Figure 1

Occurrence rates of (a) hot ($P = 1\text{--}10$ days) and (b) warm ($P = 10\text{--}100$ days) planets as a function of host star $[\text{Fe}/\text{H}]$. The occurrence of almost all types of planet seems to depend at least weakly on $[\text{Fe}/\text{H}]$ and in some cases differs depending on the orbital period range covered. The colors indicate different planet sizes. Dashed lines represent data from Petigura et al. (2018) (CKS sample, 970 *Kepler* Objects of Interest, LAMOST metallicities), and solid lines represent data from Wilson et al. (2022) (1,018 *Kepler* Objects of Interest, APOGEE metallicities). The lines represent the best fit to a distribution $f_{X,P} \propto P^\alpha 10^{\beta X}$, where the period dependence has been integrated over.

Jupiters: planets between ~ 8.0 and $24.0 R_\oplus$ and between ~ 0.1 and $10 M_{\text{Jup}}$; those with $P < 10$ days and/or $T_{\text{eq}} \gtrsim 1,000$ K are known as hot, and those with dayside temperatures $\gtrsim 2,200$ K are known as ultrahot

Sub-Saturns: planets between ~ 4.0 and $8.0 R_\oplus$, generally with masses between ~ 10 and $100 M_\oplus$ increasing with radius, at $P \leq 100$ and likely longer

2.1.1. Gas giant planets. The trend in occurrence of giant planets (Jupiters to sub-Saturns) with host star $[\text{Fe}/\text{H}]$ has been extensively studied to extract additional information and refine formation models. It is actually quite a strong function, often parameterized as a power law with an index around two (Fischer & Valenti 2005, Sousa et al. 2011; see also Mortier et al. 2013) (Figure 1). Furthermore, it appears that not only the presence of close-in giant planets but also their masses (Sousa et al. 2008, Ghezzi et al. 2010a) and radii (Buchhave et al. 2014, Schlaufman 2015, Narang et al. 2018, Petigura et al. 2018) are positively correlated with the host star metal content, although how the bulk metal content of giant planets depends on stellar metallicity is less clear (see Section 4.1; Thorngren et al. 2016, Teske et al. 2019). On the basis of data mostly from FGK dwarfs, the “break point” in planet mass, when the occurrence of giant planets no longer appears to depend on host star metallicity, is somewhere between 4 and $10 M_{\text{Jup}}$ (Santos et al. 2017, Narang et al. 2018, Schlaufman 2018, Maldonado et al. 2019), even when differences in host star masses and thus disk masses are accounted for, suggesting that planets above this mass range may form differently, perhaps via gravitational instability. This observational transition provides an independent distinction between a planet and a star that does not depend on an object’s composition or internal structure, and also helps constrain the properties (viscosity and initial mass) of disks forming planets via core accretion such that they must not form planets larger than $10 M_{\text{Jup}}$ (Schlaufman 2018). A trend of decreasing host star metallicity with increasing planet mass is also observed in a sample of 22 longer-period ($P \gtrsim 1$ year), directly imaged giant planets, although in the currently small sample the median mass is $\sim 12.5 M_{\text{Jup}}$, with the lowest-mass object at $2 M_{\text{Jup}}$, and the range of stellar metallicities is wide (spanning -0.65 to 0.30 dex) (Swastik et al. 2021). Brewer et al. (2018) confirmed that the metallicity trend for cool giant planets ($M_p > 0.5 M_{\text{Jup}}$ or $R_p > 0.75 R_{\text{Jup}}$ at > 0.3 AU), detected mostly through radial velocities, tracks

that of hot giant planets for $[\text{Fe}/\text{H}] \gtrsim -0.3$ dex, although there is a hint of an increase in the fraction of planet systems hosting a cool Jupiter planet below this value.

The metallicity–period–eccentricity distribution of giant planets has also been suggested to reveal clues about their formation. Dawson & Murray-Clay (2013) set out to investigate the processes that deliver gas giant planets to their short orbital periods (~ 10 – 100 days, ~ 0.1 – 1 AU) from farther out in the disk, beyond the ice line, where they most likely formed (e.g., Lin & Ida 1997, Ida & Lin 2008, Kennedy & Kenyon 2008). These processes fall into two basic categories: migration smoothly through a disk that still contains substantial gas or migration more violently through gravitational interactions with stars or sibling planets (see Dawson & Johnson 2018 and references therein). Dawson & Murray-Clay (2013) pointed out a statistically significant difference in the eccentricity distributions of radial-velocity-detected giant planets orbiting $[\text{Fe}/\text{H}] \geq 0$ versus $[\text{Fe}/\text{H}] < 0$ stars, such that those orbiting the more metal-rich stars have a wider range of eccentricities. Additionally, they found that eccentric proto-hot Jupiters experiencing tidal circularization also orbit primarily $[\text{Fe}/\text{H}] > 0$ stars and that there are more short-period giant planets around metal-rich versus metal-poor stars. They concluded that planet–planet scattering is likely the culprit for the orbits of close-in giant planets around metal-rich stars, around which multiple planets are more likely to form and lead to dynamical interactions/instability, whereas smooth disk migration does not have the metal-rich star requirement. However, Yee & Winn (2023) recently used a more homogeneous and complete sample of transiting hot Jupiter ($P < 10$ days) exoplanets to demonstrate that the period distributions of metal-rich versus metal-poor host stars do not show any difference. Their constraint on the lack of a detectable difference suggests that one planet delivery pathway [from the prescription of Nelson et al. (2017), disk migration or high-eccentricity migration] dominates at the $\geq 70\%$ level, at least for the closest-in giant planets. However, Yee & Winn (2023) did not investigate planet eccentricity distributions in their sample, and they suggest that a new joint period–eccentricity analysis is warranted.

If, broadly speaking, giant planet formation (via core accretion) is increasingly likely with increasing host star metallicity, is there some cutoff metallicity below which giant planets will simply not form? Johnson & Li (2012) investigated how the dust-to-gas ratio affects the time it takes for dust grains to settle in the disk midplane (after which planetesimal formation is expected to be relatively rapid; e.g., Dullemond & Dominik 2005, Brauer et al. 2008) versus the disk lifetime. Since the temperature and density of the disk change the dust-settling timescale, and these parameters change with distance r from the host star, the authors found that the critical dust-to-gas ratio ($[\text{Fe}/\text{H}]_{\text{crit}}$) for planet formation depends linearly on $\log(r)$ and can be estimated with $[\text{Fe}/\text{H}]_{\text{crit}} \simeq -1.5 + \log(r/1 \text{ AU})$. Any planet systems found below this relation would be candidates for an alternative mode of formation besides core accretion, although Johnson & Li (2012) note several important assumptions that would not rule out core accretion for a planet in their “forbidden zone.”

How do observations compare to this prediction? Radial velocity studies of metal-poor solar-type stars, accounting for detection limits, suggest a lower limit of giant planet (~ 1 – $4 M_{\text{Jup}}$; $P < 3$ years) formation around $[\text{Fe}/\text{H}] = -0.7$ dex (Sozzetti et al. 2009, Mortier et al. 2012), although they also suggest a flattening out of the occurrence rate–metallicity relation below $[\text{Fe}/\text{H}] = 0$. Recently, the *Transiting Exoplanet Survey Satellite* (TESS) (Ricker et al. 2015) provided coverage of a wide range of stellar types and $\sim 85\%$ of the sky, allowing Boley et al. (2021) to constrain the occurrence rate of 0.8 – $2.0 R_{\text{Jup}}$, 0.5 – 10 day planets around $-2.0 \leq [\text{Fe}/\text{H}] \leq -0.6$ dex halo stars to be 0.04 – 0.36% , with a mean upper limit of 0.18% . The authors proposed a similar lower $[\text{Fe}/\text{H}]$ limit as the radial velocity surveys, between -0.7 and -0.6 dex for hot Jupiter formation. Currently, the most metal-poor star known to host a hot Jupiter has $[\text{Fe}/\text{H}] = -0.6 \pm 0.19$ dex (WASP-98; Hellier et al. 2014). Several giant planets have been reported at

longer orbital periods with lower $[\text{Fe}/\text{H}]$ values; the lowest, around $[\text{Fe}/\text{H}] \sim -0.7$ dex, are 24 Boo b (Takarada et al. 2018) and HD 11755 b (Lee et al. 2015), although the former is around an evolved giant star. All these planets are still well above the forbidden zone predicted by Johnson & Li (2012). The metallicity limits of small planet formation are discussed further in Section 2.1.2, and Section 5 presents future directions in the search for planets around metal-poor stars.

2.1.2. Smaller planets. Unlike short-period giant planets, the occurrence of smaller ($\lesssim 4 R_{\oplus}$) and less massive ($\lesssim 20 M_{\oplus}$) planets generally has a weaker to nonexistent (depending on size) dependence on host star metallicity (Sousa et al. 2008; Ghezzi et al. 2010a, 2021; Buchhave et al. 2012, 2014; Schlaufman 2015; Zhu et al. 2016; Petigura et al. 2018) (**Figure 1**). Indeed, it is the probable host star to two $M_p \sin i = 5\text{--}7 M_{\oplus}$ planets that currently holds the lowest metallicity record at $[\text{Fe}/\text{H}] \sim -0.89$ dex (Anglada-Escudé et al. 2014). This weak metallicity dependence is predicted by core accretion models, although different model prescriptions can change the expected number of small planets as a function of host star metallicity (Matsumura et al. 2017) and/or the metallicity at which the frequency of small/low-mass planets may start to decrease due to the presence of giant planets (Hasegawa & Pudritz 2014). As discussed by Zhu (2019), it is important to distinguish between the fraction of stars with a planet and the average number of planets per star; this distinction becomes especially relevant with small planets because they are often found in multiplanet systems (e.g., Zhu et al. 2018, He et al. 2021; for a review, see Weiss et al. 2023). Zhu (2019) outlines tracers for each of these quantities and compares them for planets in the *Kepler*-LAMOST sample,² finding that the occurrence of small planet systems increases slightly with host star $[\text{Fe}/\text{H}]$ (from -0.2 to 0.2 dex; a factor of 4.5 less than giant planets) but that the number of small planets per star does not depend as strongly on $[\text{Fe}/\text{H}]$ and may reach a plateau around $[\text{Fe}/\text{H}] = 0.1$ dex. However, when systems with at least four transiting planets are excluded, the plateau disappears—these high-multiplicity planet systems are concentrated in the range $0 \leq [\text{Fe}/\text{H}] \leq 0.1$ dex, so they bias that bin in metallicity—and both the fraction of stars with planets and the average number of planets per star increase with stellar metallicity.

This finding is related to a key question in planet formation: How do outer giant planets affect inner smaller planets? Several studies have found that cold Jupiter planets appear (of order threefold) more often in systems that also host inner small planets (Zhu & Wu 2018, Bryan et al. 2019, Herman et al. 2019; these studies do not all use the same giant/outer and small/inner planet definitions, but they find qualitatively consistent results). In particular, Zhu & Wu (2018) found a rise in the frequency of cold giant planets in systems with inner small planets around $[\text{Fe}/\text{H}] > 0.1$ dex stars and a lack of systems with four or more transiting planets around $[\text{Fe}/\text{H}] > 0.2$ dex stars. Brewer et al. (2018) also found that the fraction of host stars of compact multiplanet systems (with three or more planets in orbits < 1 AU) is roughly flat in the range $-0.3 < [\text{Fe}/\text{H}] < 0.4$ dex but increases sharply below that range, where giant planets are infrequently found. A giant planet can destabilize a packed multiplanet system, reducing the number of planets per star but not the number of planetary systems. Zhu (2019) suggests that their occurrence rate–metallicity results support the idea that giant planets reduce the frequency of high-multiplicity small planets.³ Rodríguez Martínez et al. (2023) also recently found that M dwarfs hosting multiplanet systems

²The *Large Sky Area Multi-Object Spectroscopic Telescope* (LAMOST) surveyed more than 30% of all *Kepler* stars as of 2017, agnostic to whether or not they hosted planets (for sample details, see Zhu et al. 2018).

³Recently, Zhu (2024) pointed out that it is metal-rich super-Earth systems that have an excess of cold Jupiters; that is, the conditional rate $P(\text{CJ}|\text{SE}; [\text{Fe}/\text{H}] > 0)$ is consistently $\sim 40\%$ across different samples versus a $P(\text{CJ}; [\text{Fe}/\text{H}] > 0)$ value $\lesssim 20\%$. With a larger sample, Bryan & Lee (2024) more recently confirmed a 2.7σ positive correlation between cold gas giants and super-Earths in $[\text{Fe}/\text{H}]_* > 0$ systems that disappears for $[\text{Fe}/\text{H}]_* \leq 0$ systems.

are significantly more metal-poor than M dwarfs hosting single-planet systems; these low-mass stars are generally expected to lack giant planets regardless of their metallicity (although this is an active area of research; e.g., Gan et al. 2023), so an increase in multiplanet systems at low metallicity may be due to more than just the absence of giant planets.

Trends in the planet size–planet period–stellar metallicity space have also been illuminating for our understanding of small planet formation. On the basis of *Kepler* results, hot ($P \lesssim 10$ days), likely rocky ($R_p < 1.7 R_\oplus$) planets seem to preferentially orbit more metal-rich stars, whereas at longer periods, these super-Earth planets show a flat or potentially decreasing occurrence (planets per star) with increasing $[\text{Fe}/\text{H}]$ (Mulders et al. 2016, Petigura et al. 2018, Wilson et al. 2018) (**Figure 1**). While it was initially suggested that these so-called hot rocks might be hot Jupiter remnants, this seems unlikely, as the host stars of ultrashort-period planets have significantly different metallicity distributions than those of hot Jupiters, with the former being similar to stars hosting short-period sub-Neptune planets (Winn et al. 2017). Thus, it is still possible that these hot rock planets are the remaining solid cores of smaller but still gas-enveloped planets. Hot sub-Neptunes are also found around more metal-rich stars (their occurrence is just not as strongly tied to metallicity as hot Jupiters; **Figure 1**), and at longer periods they still show a slightly increasing occurrence (planets per star) with metallicity (Petigura et al. 2018, Wilson et al. 2022). Alternatively, the correlation between the occurrence of short-period small planets and their host star metallicity could indicate that close-in planet formation is more efficient around high-metallicity stars through, for instance, more efficient inward migration of solids, a closer-in dust sublimation radius, or a close-in corotation radius (where gas is accreted onto the stellar surface), although the latter two scenarios seem unlikely (Wilson et al. 2018).

Looking beyond hot rocky planets, Owen & Murray-Clay (2018) investigated the same California-*Kepler* survey sample as Petigura et al. (2018) in the context of atmospheric loss, specifically at short periods ($P < 2.5$ days), where atmospheric evaporation is expected to be very effective, and $P > 25$ days, where atmospheric evaporation should be minimal. They pointed out that (a) both short- and long-period sub-Neptunes are larger around higher-metallicity stars, (b) short-period super-Earths are also larger around higher-metallicity stars, and (c) long-period super-Earths are more common around low-metallicity stars but their radius distribution is metallicity independent. The authors provide a physically motivated framework tying together these results: The formation of super-Earth and sub-Neptune planets around higher-metallicity stars results in larger/higher-mass solid cores—at short periods, where atmospheres should be mostly stripped, all small planets are larger around higher-metallicity stars. Furthermore, increasing host star metallicity may result in increasingly metal-rich gaseous envelopes for the planets massive enough to accrete them, meaning these planets can better “resist” evaporation at shorter orbital periods versus those in lower-metallicity systems. The longer-period planets with enough mass to be able to collect larger gaseous envelopes and thus increase their size are also larger around more metal-rich stars. The longer-period, $1\text{--}1.8 R_\oplus$ planets are more frequent around more metal-poor stars, which presumably have less material available in the disk to form larger cores fast enough to accrete H/He envelopes. Thus, these planets may have originated as rocky and never accreted a significant nebular gas component, pointing to a later stage of small planet formation after the disk dispersed around more metal-poor stars. This hypothesis presents the intriguing possibility that “born rocky” planets form in significantly different chemical environments and thus may have different internal compositions (a topic revisited below in Section 4.2).

2.2. Beyond Iron Abundances

While Fe is the most easily and frequently measured element in most solar-type star spectra and a standard metric of total metallicity, as astronomers we have a plethora of metals to choose to

Sub-Neptunes:

planets between ~ 1.7 and $4.0 R_\oplus$, generally with masses between ~ 3 and $25 M_\oplus$ increasing with radius, at $P \leq 100$ days and likely longer; those with $P \leq 1$ day and $R_p \leq 2 R_\oplus$ are known as ultrashort-period planets

Super-Earths:

planets between ~ 1.0 and $1.7 R_\oplus$, generally with masses below $\sim 10 M_\oplus$, at $P \leq 100$ days and likely longer; those with $P \leq 1$ day are known as ultrashort-period planets

measure in host stars, the abundances of which may also be important for planet formation. As mentioned in Section 1, there are moderately well characterized correlations between Fe and other metals as a result of GCE. Therefore, to isolate the contribution of other metals, we can consider $[X/\text{Fe}]$, normalizing the other metal X by the star's Fe abundance. However, stellar chemistry also traces kinematics and age, so it can be difficult to determine which of these factors is most influential in planet formation. Overall, most comparisons of abundances and abundance ratios (e.g., C/O, Mg/Si) between stars that host and stars not known to host planets have found no meaningful differences that are not explained simply by GCE, regardless of planet formation [but see Robinson et al. (2006) for an example of significant enhancement of Si and Ni in planet host stars when controlling for Fe].

A significant and important finding in planet occurrence beyond just Fe host star abundances is that $[X/\text{Fe}]$ ratios for a suite of elements including Mg, Al, Si, Sc, and Ti are systematically higher at low $[\text{Fe}/\text{H}]$ (usually $\lesssim -0.3$ dex) in host stars versus stars not known to host planets (Brugamyer et al. 2011, Adibekyan et al. 2012b). While the enhancement of $[X/\text{Fe}]$ at low $[\text{Fe}/\text{H}]$ is found in host stars to Jupiters, Neptunes, and super-Earths, the overall frequency of larger/more massive planets decreases with $[\text{Fe}/\text{H}]$, so the trend is dominated by the smaller planets (Adibekyan et al. 2012a).⁴ Notably, this $[X/\text{Fe}]$ enhancement does not appear to be the case for O, even though it is an α element like Si, is incorporated into many rock-forming minerals, and is the major constituent of Earth's crust. This finding may indicate that, particularly in Fe-poor environments, it is the initial grain nucleation (perhaps more related to the Si abundance) versus growth by accretion of ices (perhaps more related to the O abundance) that serves as the limiting step to planet formation (Robinson et al. 2006, Brugamyer et al. 2011) and/or that the solid compositions of these planets will be significantly different. Planet host stars are also not preferentially enhanced in other elements typically associated with volatile species like C and N (Suárez-Andrés et al. 2016, 2017), although Wilson et al. (2022) found a statistically significant trend in $[\text{S}/\text{Fe}]$ with both planet radius and period in their analysis of the APOGEE-KOI program. This trend cannot be explained by other confounding correlations, and its source remains unknown.

The fact that planet host stars do not stand out in many stellar abundance contexts does not mean that there is a lack of variety in their compositions. Indeed, especially for small planets, the range of chemical environments for planet formation spans the entire Galaxy, and perhaps beyond, if some small percentage of stars currently within the Milky Way actually originated in a neighboring dwarf galaxy (e.g., Newton et al. 2021). The implications of this chemical variety on planet compositions are explored further in Section 4.

3. ANOMALOUS ABUNDANCES

Within the study of host star abundances, a few specific populations stand out as particularly useful and interesting for isolating the effect of planet formation and, thus, for better pinpointing where it occurred and whether or not planets have been detected there (yet).

3.1. Solar Twins and Siblings

While the giant planet–metallicity correlation (Section 2.1) was identified early on in the history of exoplanet studies, more subtle stellar abundance signatures related to planets, particularly small rocky ones, came with increasing planet detections as well as more precise approaches for measuring stellar abundance differences. As discussed in Section 1, the highest-precision stellar abundances—and, thus, sensitivity to the smallest differences—come from comparing similar

⁴However, Maldonado et al. (2018) report that cool Jupiter host stars show an enhancement in α elements in comparison to stars hosting hot Jupiters.

stars differentially, since systematic errors in model atmospheres and/or line parameters effectively cancel out. The star we know best and can thus make the most accurate comparisons with just so happens to have formed rocky planets. So, it makes sense to investigate whether the Sun's abundances offer a signature of rocky planet formation, which could then help narrow down the number of stars around which we should spend significant time looking for similar planetary systems.

The first promising evidence of a main-sequence stellar abundance signature related to rocky planet formation⁵ was presented by Meléndez et al. (2009), who conducted a differential spectroscopic study of 11 solar twins (Cayrel de Strobel 1996) using $R \sim 65,000$, $S/N \sim 400$, and 340–1,000 nm spectra, all collected with the same instrument. The parameters of the spectra like resolution and S/N are important because they can influence the systematic errors that can arise from subtle mismatches when using spectra from different instruments or configurations (Bedell et al. 2014). Meléndez et al. measured abundances with very small internal errors around 0.01 dex and showed that the Sun is preferentially enhanced in volatile elements (low condensation temperature, T_{cond}) but depleted in refractory elements (high T_{cond}) compared with the solar twins (**Figure 2**). The authors suggested that this abundance– T_{cond} trend in the Sun was indeed a beacon for rocky planet formation—the refractories are “missing” from the Sun because they were incorporated into the terrestrial planets—which Chambers (2010) confirmed could be matched by adding $4 M_{\oplus}$ of a mixture of Earth-like and carbonaceous chondrite-like material to the present-day solar convection zone.

As exciting as this suggestion is, there are several alternative explanations and additional factors to consider. Some were pointed out by Meléndez et al. (2009) themselves and others were identified later, as our knowledge of exoplanets grew. First, while the comparison stars were solar twins, they were not necessarily formed at the same time or location as the Sun, so they could have experienced different effects of GCE that influenced their $[X/\text{Fe}]$ – T_{cond} trends (e.g., Adibekyan et al. 2014, Nissen 2015, Spina et al. 2016). When this GCE influence, a linear trend in $[X/\text{Fe}]$ with age, is fitted and removed, the Sun may still be anomalous relative to the majority of its twins—Bedell et al. (2018) found that the Sun is more deficient in refractories than 82% of a larger sample of 68 solar twins corrected for GCE effects (but see the different estimates of GCE effects in Cowley & Yüce 2022). Note, however, that it is impossible to tell the difference between refractory depletion in the Sun and refractory enhancement in other comparison stars, as their primordial compositions are unknown. It has also been suggested that the Sun's refractory depletion signature could arise if the gas from which it formed was initially cleansed of dust by radiation pressure from nearby massive stars (Gustafsson et al. 2010, Gustafsson 2018a,b).

Second, it is important to consider the timing of any event leading to an enhancement or depletion signature in stellar abundances in relation to the size of the stellar convection zone. The size of the young ($\lesssim 10$ Myr) Sun's convection zone was likely 50–100% by mass (Kunitomo et al. 2018) versus 2.5% today, which means that more like 80–160 $M_{\oplus}$ of rocky material would be needed to produce the Sun-solar twin difference if it arose during the first ~ 10 Myr, much more than is bound in Solar System rocky bodies. The mass of solids in the larger planets in the Solar System could reach the lower values in this range, but given the high fraction of volatile-rich ices in these solids, their missing mass from the Sun does not obviously produce the refractory depletion trend. However, as discussed below in Section 3.2, planet engulfment is a feasible alternative, assuming that it occurs at later stages when less material would be required to produce an abundance change in the stellar photosphere.

Solar twins: stars with T_{eff} within 100 K of solar, surface gravity ($\log g$) within 0.1 dex of solar, and metallicity ($[\text{Fe}/\text{H}]$) within 0.1 dex of solar

⁵But see Section 4.2.1—abundance evidence of rocky planetary debris in-falling on white dwarfs was detected earlier.

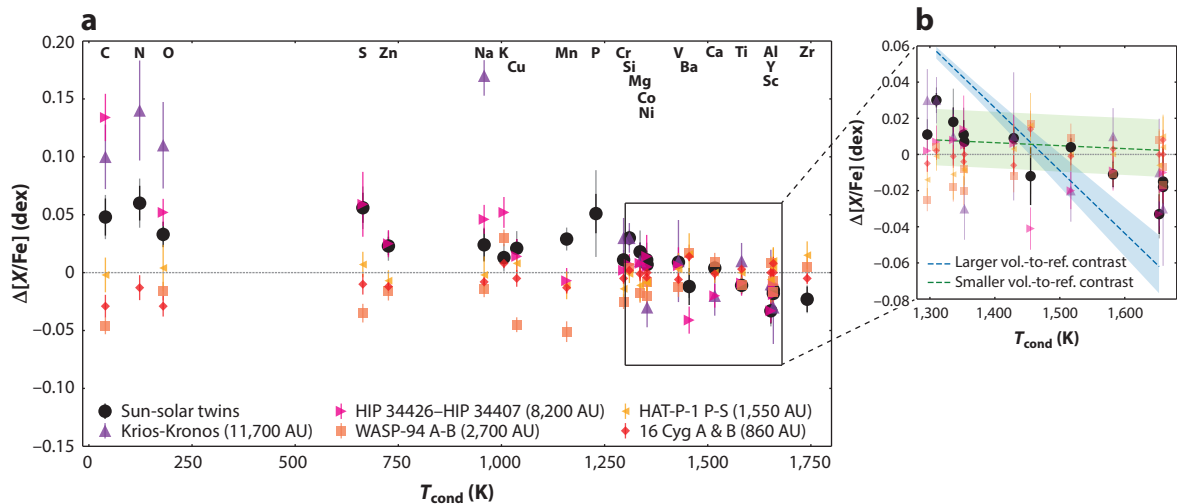


Figure 2

(a) Trends in differential stellar abundances versus condensation temperature (50% in solar composition gas) for various twin binary systems (*colored symbols*) and for the Sun minus solar twins (*black points*). Individual elements are listed at the top, and the horizontal gray dashed line indicates zero difference. The figure illustrates the challenge of interpreting abundance– T_{cond} trends. The precision on these abundances is extremely good, and yet not all systems show an obvious trend like the refractory depletion observed in the Sun, although the $\Delta[X/\text{Fe}]$ values generally decrease overall with decreasing separation between the binary pair. (b) Zoomed-in view of the T_{cond} range investigated by Nibauer et al. (2021), plotting the two distinct trends found in a sample of $\sim 1,800$ solar analogs from APOGEE-2. Most of the stars show a small contrast in the ratio of volatile to refractory abundances (*green dashed line, shaded uncertainty*), while 10–30% show a greater contrast between volatile and refractory abundances (*blue dashed line, shaded uncertainty*). The Sun belongs to the more frequently occurring small-contrast population (*green*), suggesting that it may not be so anomalous when compared with other solar analogs. In all of the binary star cases, the figure shows (hotter–colder star) abundances for consistency because, in some cases, both stars host planets. When the tabulated measurement is $[X/\text{H}]$, it is normalized by each star's $[\text{Fe}/\text{H}]$ to get $[X/\text{Fe}]$. The neutral species abundance is taken when both neutral and singly ionized are listed, and the atomic abundances are taken when both atomic and molecular are listed. Li abundances are not plotted; these are usually measured by synthesis fitting to an individual line. The gray/black error bars for the Sun-solar twins include observational errors both in the Sun and solar twins and in the solar twins alone. Abbreviation: P-S, primary–secondary. Panel *a* adapted with permission from Meléndez et al. (2009) (Sun-solar twins); copyright 2009 AAS, and data also taken from Oh et al. (2018) (Kronos-Krios), Ramírez et al. (2019) (HIP 34426–HIP 34407), Teske et al. (2016a) (WASP-94 A-B), Maia et al. (2019) (16 Cyg A & B), Liu et al. (2014) (HAT-P-1 P-S), and Nibauer et al. (2021) (small/large contrast).

Third, we now know that small planets occur relatively frequently and giant planets much less so (e.g., Hsu et al. 2019, Wittenmyer et al. 2020). So, it is hard to reconcile the fact that a large fraction of Sun-like stars would lack a similar signature of rocky planet formation. Also, multiple studies have been unable to confirm the same rocky planet $[X/\text{Fe}]$ – T_{cond} signature in Sun-like (albeit not twin) stars known to host close-in rocky planets (González Hernández et al. 2010, 2013; Schuler et al. 2011b, 2015; Liu et al. 2020). Instead, perhaps, the $[X/\text{Fe}]$ – T_{cond} trend in the Sun relative to solar twins is a beacon of giant planet formation. As described by Booth & Owen (2020), when giant planets form early and open gaps in the protoplanetary disk, this process can preferentially trap dust while allowing refractory-poor gas to flow through toward the star. These authors modeled the evolution of protoplanetary disks surrounding young stars and matched the observed ~ 0.04 dex refractory depletion observed in the Sun, although they did not attempt to reproduce the specific trend with T_{cond} . Additional evidence supports such preferential refractory material trapping from observations of Herbig Ae stars, young A stars with very thin, if any, convection zones at their surfaces, allowing recently accreted material to dominate the photospheric composition (Jermyn & Kama 2018). A comparison of Herbig Ae stars hosting cold,

Solar analogs:

stars with similar parameters to the Sun (e.g., ± 500 K, ± 0.3 dex in $[\text{Fe}/\text{H}]$) but not as similar as solar twins; could be yet-to-be-confirmed solar twins requiring better parameter determination

flat disks with stars hosting transitional disks with large inner cavities depleted in millimeter-sized dust grains shows that the second group is depleted by a factor of ~ 10 in Fe, Mg, and Si but not in C or O (Kama et al. 2015). Further research also showed depletion in Ti and the intermediate T_{cond} element S (Kama et al. 2019). Additional studies of stars hosting distant gas giants could help shed light on this hypothesis of an $[X/\text{Fe}]$ – T_{cond} trend indicating giant planet formation.

While studies searching for refractory depletion in solar twins have focused mostly on relatively small samples with high resolution and S/N spectra, Nibauer et al. (2021) applied a likelihood-based approach including a hierarchical mixture model to the less-precise abundances of 1,800 solar analogs in the APOGEE-2 catalog (DR16; Jönsson et al. 2020) to assess whether there is evidence for two populations of stars (depleted and not depleted in refractories). The authors focused on the most precisely determined APOGEE-2 abundances for dwarf stars and measured only the $[X/\text{Fe}]$ – T_{cond} trend across a limited range (Si, Mg, Ni, Ca, and Al, covering $T_c \sim 1,300$ – $1,650$ K). They indeed found evidence for two populations; the not-depleted population had a significant slope⁶ across T_{cond} (**Figure 2b**). Interestingly, they found that $\sim 85\%$ of stars are in their depleted population, with an approximately flat slope across $T_{\text{cond}} = 1,300$ – $1,650$ K (**Figure 2b**) and that while the Sun is still more depleted than $\sim 80\%$ of their full sample, it still falls in the middle of a distribution of stars with similar $[X/\text{Fe}]$ – T_{cond} trends. This finding calls into question just how anomalous the Sun and/or some level of refractory-to-volatile depletion is among stars fairly nearby in the Galaxy. Similarly, Behrard et al. (2023b) found that APOGEE-2 (DR17) abundances of *Kepler* planet host stars and matching doppelgänger stars (with the same fundamental parameters but no knowledge of planets) have indistinguishable T_{cond} trends—there is nothing special about the planet hosts—although the authors note that they are limited in part by abundance precision.

3.2. Binary Stars

A natural extension of studying stars that are twins of the Sun is studying stars that are twins of one another—binary stars that formed in the same place at the same time and are very similar in mass, metallicity, and temperature. Spectra of these stars can be analyzed via the same strictly differential technique that minimizes systematic errors and thus results in high-precision abundances, which are not subject to different influences of GCE. Thus, differences in otherwise identical binary star abundances could be related to planet formation. Extending this idea even further, the pattern of differences in the abundances might reveal something about the type of planet formation that occurred (e.g., inside or outside the water ice line; Bitsch et al. 2018).

At first, the giant planet–metallicity correlation (Section 2.1) was thought to potentially be caused by the pollution of the stellar convective envelope by planet or planet-forming material. This hypothesis motivated astronomers to search for large abundance differences in wide binary systems. Initial line-by-line differential studies of moderate samples (tens of stars; Gratton et al. 2001, Desidera et al. 2004, 2006) found a few outlying cases of large ($\gtrsim 0.1$ dex) abundance differences but indicated that it is uncommon for stars to ingest enough material to cause a significant difference and, thus, the clear enhancement in metallicity among giant planet host stars.⁷ As discussed below, planet engulfment may indeed be happening, but its impact on stellar abundances is predicted to be more subtle.

Solar siblings: stars that formed from the same stellar cluster at the same time as the Sun but may not still be close in distance to the Sun today; they have similar composition and age but not necessarily other parameters

⁶Note that, as usual, $[X/\text{Fe}]$ as used by Nibauer et al. (2021) refers to star-minus-Sun differential abundances. **Figure 2** uses the term Sun-solar twins to match the original figure from Meléndez et al. (2009), meaning that the trends shown in **Figure 2b** are flipped in slope versus what is reported by Nibauer et al. (2021).

⁷However, Desidera et al. (2004) diligently pointed out that the significance of their conclusion decreases if the occurrence rate of planets around binary stars and/or in clusters is less than half of that around normal

Interest in studying binary stars, particularly twins, was renewed by the solar twin studies discussed above, as well as by increasing numbers of exoplanet detections. It became possible to compare two stars in a binary system in which one was known to host one or more planets, or in which the stars were known to host different types of planets, providing additional context for measured abundance differences. **Table 1** presents a selection of detailed binary star abundance differences studies, and **Figure 2** plots a subset. Perhaps the most popular example is 16 Cyg A & B, a pair of stars that themselves are solar twins, wherein B hosts an eccentric super-Jupiter planet at nearly 800 days. While many authors have measured the pair and found the A component (which is slightly hotter) to be enriched in metals by ~ 0.04 dex, whether or not there is a trend in $\Delta[X/\text{Fe}]$ with T_{cond} has been debated (Laws & Gonzalez 2001, Schuler et al. 2011a, Tucci Maia et al. 2014, Nissen et al. 2017, Maia et al. 2019). The latest results from the highest S/N spectra indicate a slope in $\Delta[X/\text{H}]$ (A–B) versus T_{cond} of $1.56 \pm 0.24 \times 10^{-5}$ dex K^{-1} . This trend was originally attributed to the effect of the rocky core of 16 Cyg B b—the B star is preferentially depleted in refractories—but Maia et al. (2019) also found the A component to be overabundant in Li, above what is expected for its age, as well as in Be. This finding suggests instead that the A–B abundance trend is better explained by engulfment of $2.5\text{--}3 M_{\oplus}$ of Earth-like material by 16 Cyg A.

Overall, the picture of twin binary star studies is muddy, as summarized in **Table 1** and by Oh et al. (2018, appendix) and Behrard et al. (2023a, table 1). Different combinations of systems with/without planets, with/without abundance differences, and with/without trends with T_{cond} exist in the literature. Most abundance differences between twin binaries, when detected, are at the 0.05 dex level or lower, emphasizing the extreme precision in these studies (**Figure 2**). There are a few notable exceptions of more significant differences—the HD 240430/29 (Kronos-Krios; Oh et al. 2018) and HIP 34407/26 (Ramírez et al. 2019, Nagar et al. 2020) systems show differences at the 0.1–0.2 dex level—but it is challenging to draw conclusions from the majority of results, especially given that many stars may host planets that remain undetected.

Let us take a step back to answer a more fundamental question: What is the typical difference in binary star abundances? The answer has implications for chemical tagging—using the chemical similarity of stars that are no longer physically close to one another to trace them back to their birth cloud and siblings (Freeman & Bland-Hawthorn 2002)—as well as metallicity calibrations for cool M dwarf stars, where an M dwarf star is assumed to have approximately the same composition as the higher mass companion (see Montes et al. 2018 and references therein). Aside from planet formation and inhomogeneous molecular clouds, relatively small offsets in T_{eff} and $\log g$ between stars can lead to different surface abundances through different levels of internal turbulence and, hence, differential atomic diffusion (Dotter et al. 2017, Sevilla et al. 2022). The high-precision astrometry from *Gaia* (Gaia Collab. 2023) has made it possible to greatly expand the sample of wide binaries and comoving pairs (e.g., El-Badry & Rix 2018) with which we can assess typical binary abundance differences. It appears that within wide ($\sim 300\text{--}50,000$ AU projected separation) $\Delta T_{\text{eff}} < 200$ K binary systems, $\sim 80\%$ of pairs are homogeneous at the 0.02 dex level in $[\text{Fe}/\text{H}]$ and similarly for a range of other elements, far more similar than randomly paired field stars of similar spectral type (Hawkins et al. 2020). Going further in separation, even very similar comoving pairs ($\Delta v_{3D} < 2$ km s^{-1}) separated by $\gtrsim 10^5$ AU, which are no longer bound, still have a relatively high fraction ($\sim 70\%$) of homogeneity versus random pairings, as least for $[\text{Fe}/\text{H}]$ [standard deviation of 0.08 versus 0.23 dex (Nelson et al. 2021) or 0.109 versus 0.215 dex (Yong et al. 2023)]. The larger sample used by Yong et al. (2023) shows a natural break point in the

field stars. Moe & Kratter (2021) later determined that wide binaries with separations greater than 200 AU do not significantly affect planet formation.

Table 1 Selection of detailed abundance studies of binary stars

System	$T_{\text{eff,avg}}$ (K)	$ \Delta T_{\text{eff}} $ (K)	$\log g_{\text{avg}}$	$ \Delta \log g $	$[\text{Fe}/\text{H}]_{\text{avg}}$ (dex)	$ \Delta [\text{Fe}/\text{H}] $ (dex)	Separation (AU)	Reference(s)	Comments
16 Cygni A & B	5,798	69	4.33	0.05	0.083	0.04	860	Maia et al. (2019) ^a Laws & Gonzalez (2001) Schuler et al. (2011a) Ramírez et al. (2011)	B hosts eccentric giant planet, T_{cond} trend debated
XO-2 N & S	5,470	60	4.44	0.02	0.374	0.053	4,500	Ramírez et al. (2015) ^a Teske et al. (2013) Biazzo et al. (2015) Biazzo et al. (2015)	N hosts hot Jupiter, S hosts two longer-period giant planets, possible T_{cond} trend
HD 20781 HD 20782	5,557	465	4.46	0.10	0.010	0.060	9,000	Mack et al. (2014)	20781 hosts 2 mod e Neptune-mass planets within 0.3 AU, 20782 hosts Jupiter-mass planet with $e = 0.97$ at 1.4 AU
HAT-P-1 + secondary	6,150	202	4.40	0.07	0.151	0.009	1,550	Liu et al. (2014)	Cooler secondary hosts transiting giant planet, no T_{cond} trend
HAT-P-4 + secondary	6,037	1	4.36	0.05	0.226	0.105	28,446	Saffe et al. (2017)	Primary hosts transiting hot Jupiter, enhanced in Li, T_{cond} trend
HD 80606 HD 80607	5,587	52	4.45	0.04	0.350	0.000	1,200	Mack et al. (2016) ^a Liu et al. (2018)	80606 hosts $e = 0.94$ giant planet, no T_{cond} trend
WASP-94 A & B	6,155	86	4.19	0.08	0.312	0.013	2,700	Teske et al. (2016a)	A hosts transiting misaligned hot Jupiter, B hosts aligned hot Jupiter, slight T_{cond} trend
ζ^1/ζ^2 Reticuli	5,782	144	4.54	0.01	-0.205	0.020	3,713	Saffe et al. (2016)	ζ^2 hosts debris disk, T_{cond} trend
HD 240430 HD 240429	5,841	75	4.38	0.10	0.105	0.190	11,704	Oh et al. (2018)	Kronos-Krios, more metal-rich has high Li, strong T_{cond} trend
HD 134439 HD 134440	5,015	138	4.67	0.02	-1.410	0.040	8,617	Reggiani & Meléndez (2018)	134440 low in Li and enhanced [X/H], no T_{cond} trend
HIP 34407 HIP 34426	5,982	51	4.33	0.01	-0.456	0.177	8,200	Ramírez et al. (2019) ^a Nagar et al. (2020)	Strong T_{cond} trend HIP 34426 more depleted except in Li
HD 133131 A & B	5,697	13	4.23	0.02	-0.355	0.004	371	Liu et al. (2021) ^a Teske et al. (2016b)	A hosts two eccentric giant planets at ~ 1.4 and ~ 5 AU, B hosts giant planet at ~ 6.5 AU, no obvious T_{cond} trend
HD 106515 A & B	5,267	68	4.29	0.02	0.008	0.008	233	Liu et al. (2021) ^a Saffe et al. (2019)	A hosts $\sim 9 M_{\text{Jup}}$, $e \sim 0.6$, 9.9 years period planet, no T_{cond} trend

^aSource used for parameters in table.

fraction of chemically homogeneous comoving pairs at 10^6 AU—at closer separations, $\sim 70\%$ of their pairs have $|\Delta[\text{Fe}/\text{H}]|/\sigma[\text{Fe}/\text{H}] \leq 3.0$ and $|\Delta[\text{Fe}/\text{H}]| \leq 0.04$ dex, versus only $\sim 18\%$ at larger separations.

Given that $\sim 20\text{--}35\%$ of binary stars may be chemically distinct at an above-average level, does this mean that all of these systems experienced planet engulfment? Spina et al. (2021) combined 31 new binary pair measurements with 76 from the literature in a meta-analysis across a wide temperature range ($\sim 5,000\text{--}6,450$ K) and found a higher fraction of chemically distinct pairs [defined as $|\Delta[\text{Fe}/\text{H}]| > 2 \times \max(\sigma_{\Delta[\text{Fe}/\text{H}]}, 0.01)$] with increasing $T_{\text{eff, avg}}$ (average of the two stars), which is in line with predictions from planet engulfment events, since hotter stars have thinner convection zones that can be contaminated more easily. This result was recently bolstered by Yong et al. (2023), who included a more homogeneous analysis (all abundances measured in the same way, line by line) and a larger sample (125 comoving pairs) and found a similar trend of increasing chemical inhomogeneity with increasing $T_{\text{eff, avg}}$. However, not all of the pairs in these studies were “twins”—Spina et al. (2021) required the pairs to have $\Delta T_{\text{eff}} < 600$ K and $\Delta \log g < 0.6$ dex, and Yong et al. (2023) imposed $|\Delta(BP - RP)| < 0.15$ mag and $|\Delta M_G| < 1$ mag (resulting in $\Delta T_{\text{eff}} \leq 561$ K and $\Delta \log g < 0.4$ dex). As mentioned above, different stellar evolutionary states can contribute to abundance offsets from the intrinsic effects of atomic diffusion, unrelated to planet engulfment (Liu et al. 2019, 2021; Souto et al. 2019).

Behrard et al. (2023a) recently addressed the planet engulfment question with MESA (Modules for Experiments in Stellar Astrophysics) (Paxton et al. 2011), building on past research (Vauclair 2004, Théado & Vauclair 2012) to model how planet engulfment signatures are influenced by a variety of stellar mixing processes, including convection, thermohaline mixing,⁸ diffusion, gravitational settling, and radiative levitation. Their models tested the impact of the accretion of $1\text{--}50 M_{\oplus}$ of bulk Earth composition material onto $0.5\text{--}1.2 M_{\odot}$ stars at very early-stage [reaching zero-age main sequence (ZAMS)], late heavy bombardment (500 Myr), disk accretion (10 Myr), and late-stage (300 Myr–3 Gyr post-ZAMS) times; the length of time for accretion also varied depending on the time at which it began. At later times, the convection envelopes of Sun-like stars are smaller and thermohaline mixing decreases, both potentially increasing the detectability of an accretion signature (defined by Behrard et al. as > 0.05 dex). Subsolar metallicity models also result in more detectable signatures due to the larger contrast of refractory enrichment above the background metallicity and the stars’ smaller convective envelopes; more massive stars also have smaller convective envelopes. Across the range of engulfment scenarios, most do not produce observable abundance difference signatures. The most promising candidates are stars older than 1.5 Gyr with $1.1\text{--}1.2 M_{\odot}$ masses, particularly those with low metallicity, but even in these cases the engulfment would need to occur well after ZAMS. Behrard et al. (2023a) also applied this planet engulfment model to their own *Keck*-HIRES measurements of 36 multistar systems in which at least one star is known to host a planet. They found only a single case (HAT-P-4) that met their criteria for a possible engulfment signature (including matching a model with $\geq 10 M_{\oplus}$ of engulfed mass; Behrard et al. 2023c). However, the engulfment must have happened late in the star’s lifetime, within the last 2 Gyr, and the large projected separation of the components ($\sim 30,000$ AU) (Table 1) suggests that the abundance differences could simply indicate an inhomogeneous, patchy composition of the birth molecular cloud. So, while $\gtrsim 0.02$ dex abundance differences between binary stars may occur in something like a quarter of systems,

⁸This is a form of double-diffusive instability in which a temperature gradient stabilizes an inverse mean-molecular weight gradient, characterized by “metallic fingers” of heavier material that stretch into a cooler medium to exchange heat excess. A kitchen experiment version is available at <https://www.stellarphysics.org/thermohaline-mixing>.

planet engulfment is unlikely to be the culprit in most cases [closer to $\sim 3\%$ (Behmard et al. 2023c) or $\sim 8\%$ (Liu et al. 2024)]. Also, it is really only in $\Delta T_{\text{eff}} \lesssim 200$ K systems that reliable engulfment signatures can be detected, as a result of refractory depletion rates varying with stellar type.

3.3. Lithium

Lithium abundances are a challenging but potentially powerful tracer of the physical processes occurring in stellar interiors. The Li that a star is born with is destroyed at relatively low temperatures ($\sim 2.5 \times 10^6$ K) in stellar interiors by proton-capture reactions, primarily in the pre-main-sequence stage. Broadly speaking, for FGKM dwarfs, cooler/less massive stars have lower Li than hotter/more massive stars because of their deeper convective zone envelopes, which bring material down to the temperature required to burn Li; by the end of the main sequence, lower-mass stars can be depleted by factors of 100 from the original abundance. Standard solar interior models that consider only convective mixing do not predict Li depletion in solar-temperature stars, and inferences from helioseismology indicate that the base of the Sun's present day convection zone envelope is just cool enough to avoid burning Li (Christensen-Dalsgaard et al. 1991), so the Sun should not show Li depletion at the surface. And yet, the Sun itself is depleted in Li by more than two orders of magnitude in comparison to the meteoritic values that benchmark Solar System formation (Lodders 2003), and within the solar neighborhood, Li abundances vary widely across stars with similar properties (Lambert & Reddy 2004). Furthermore, while most cooler stars (with deeper convective envelopes) indeed show significant Li depletion, even some stars hotter than the Sun (with shallower convective envelopes) still exhibit Li abundances lower than traditional models. Li abundances also depend on stellar metallicity, age, rotation, and activity level (Ghezzi et al. 2010b, Ramírez et al. 2012, Carlos et al. 2019, Martos et al. 2023), including in the open cluster M67 (Carlos et al. 2020).

Thus, evidence suggests that additional mixing must be operating inside solar-type stars to destroy Li, meaning that Li abundances offer an opportunity to understand these mechanisms that are otherwise invisible from the surface. Many such mechanisms have been proposed—rotational mixing (Pinsonneault et al. 1992, Eggenberger et al. 2010), overshooting mixing (Xiong & Deng 2009), internal gravity waves (Montalbán & Schatzman 2000, Charbonnel & Talon 2005), and atomic diffusion (Michaud 1986)—but which of them dominates, or which combination/in what proportion, is still an area of active research. Specifically of interest for this review, planet formation has been invoked as a way to influence stellar Li abundances. Interactions with the protoplanetary disk can produce rotational breaking in the star, contributing to differential rotation and potentially inducing hydrodynamical instabilities that can enhance Li burning (Bouvier 2008). Planetary migration can deplete Li by affecting the angular momentum of stars, creating a shear instability that produces effective mixing (Castro et al. 2009). Episodic accretion events might also affect mixing via thermohaline convection (Garaud 2011, Théado & Vauclair 2012) and/or increase the temperature at the bottom of the convective envelope and thus the effectiveness of Li depletion (Baraffe & Chabrier 2010). It is also possible that planet-related processes (accretion of sufficient planetary material, tidal orbital decay) could enhance Li if they occur late enough in the star's lifetime and deliver fresh Li after the majority of the original Li has been depleted. Li enhancement has been observed in both main-sequence and more evolved (giant) stars (for discussion, see Carlberg et al. 2012, Tayar et al. 2023), but for simplicity, this article focuses on solar-type stars.

If any of these planet-related mechanisms were causing Li depletion (or enhancement), we might expect to find that host stars are preferentially Li-poor (or -rich) versus stars without known planets. However, as described above, isolating the source of any difference in Li abundances is difficult, given the multitude of variables at play, and sample biases as well as how the abundances

were measured in different studies need to be carefully considered. Authors have been investigating whether stars with and without known planets show differences in Li abundance since not long after the first detection of exoplanets (Gonzalez & Laws 2000; Ryan 2000; Israelian et al. 2004, 2009; Baumann et al. 2010; Ghezzi et al. 2010b; Ramírez et al. 2012; Delgado Mena et al. 2014, 2015). Even as samples of high-resolution stellar spectra and exoplanets have grown, the consensus seems to still be that there are no significant differences in Li abundances between host and non-host stars, except perhaps within narrow parameter ranges, particularly around the solar effective temperature, where the Li abundances are most sensitive to small changes in mixing mechanisms. There are also one-off cases with large differences in Li abundances between binary stars (see Section 3.2) that cannot be explained even with more complicated stellar models that include additional transport mechanisms (Carlos et al. 2016, Galarza et al. 2021). Interestingly, Sevilla et al. (2022) recently showed that, with stellar models including additional mixing processes (convective overshoot mixing, thermohaline mixing, atomic diffusion), planet engulfment can indeed produce detectable Li enhancements in solar-type ($0.8\text{--}0.9 M_{\odot}$) stars but only for ~ 1 Gyr after engulfment. This finding suggests that, in a binary system, only a fairly recent planetary engulfment event can explain a large $\Delta(\text{Li})$ abundance, and the interpretation of such an offset would also need to take into account any difference in the stars' properties (primarily masses).

4. STELLAR COMPOSITION AS AN INDICATOR OF PLANET FORMATION AND COMPOSITION

Going beyond looking for correlations between host star abundances and the presence of different types of planets, we can use the stellar composition in more direct ways to investigate planet formation and composition.

4.1. Giant Planets

In the context of gas-dominated exoplanets, host star abundances have commonly been compared with planetary atmospheric compositions to constrain the formation location or pathway of the planet. Due to the complex and variable chemistry and physics of protoplanetary disks, where and when a planet accretes its constituent gas and solid components (and how much of each) will dictate its resulting composition. The focus of such studies has generally been on short-period giant planet atmospheric compositions, as they are the most easily observable and the presence of these planets so close to their host stars is still one of the greatest mysteries in exoplanet science. Migration is a critical component of formation theories to explain these short periods, but whether this happens smoothly through a protoplanetary disk that still contains substantial gas or through more dynamically active channels, like planet–planet scattering or Kozai migration, is not necessarily clear on a system-by-system basis. The former pathway is more likely to result in orbits aligned with the host star spin axis, versus highly inclined or even retrograde orbits, but other factors can influence planetary orbits and blur these populations (see Albrecht et al. 2022 and references therein). Knowing how far and through what parts of the disk a giant planet has migrated could help reveal the dynamical history of the system, including the giant planet's influence on other planets (see Section 2.1.2).

Before proceeding, I note that population studies of the metal content in both Solar System planets and giant exoplanets reveal that their envelopes are enriched relative to their host stars in elements heavier than H and He. This enrichment is more robustly measured in our Solar System planets (Atreya et al. 2018), but even for giant exoplanets (at least below $\sim 10 M_{\text{Jup}}$), the fact that their masses, radii, and ages are best matched by internal structure models with total metal content in excess of what would be in the core indicates that their envelopes must be highly enriched in metals relative to their host stars (Thorngren et al. 2016). This means that bulk

metallicity alone is probably not very discriminating in terms of constraining planet formation location. There is a very clear positive correlation between (cool giant, $T_{\text{eq}} \lesssim 1,000$ K and $20 M_{\oplus} < M_{\text{p}} < 20 M_{\text{Jup}}$) planet mass and heavy-element mass, and a negative correlation between planet mass and heavy-element enrichment relative to their parent stars (metal mass fraction in planet versus star, $Z_{\text{p}}/Z_{\text{star}}$). However, planets' residual bulk metallicity (the relative amount of metal versus that expected from the planet mass alone) does not appear to be correlated with their host star metallicity (Teske et al. 2019), which is puzzling—it suggests that more metals in a planet's feeding zone do not necessarily mean that the planet will end up more metal-rich—and in conflict with common predictions from formation models. However, as described below, more complex disk and planet formation processes are likely influencing a straightforward correlation.

The canonical comparison between host stars and giant planet atmospheres to constrain formation location is the carbon-to-oxygen (C/O) ratio, which typically is not normalized to solar (no bracket notation); the solar C/O_⊙ ratio is ~ 0.5 (Asplund et al. 2009). In principle, this ratio is feasible to measure in stars, disks, and planet atmospheres, and it encompasses key elements in planet-forming materials. Whether the key elements are in gaseous or solid form depends on the location in the disk, with increasing distance/decreasing temperature generally leading to the snow-line sequence of H₂O around 2 AU, CO₂ around 10 AU, CO around 40 AU, and CH₄ around 100 AU (in a static model of a disk around a young solar-type star; e.g., Öberg et al. 2011) (see also **Figure 3c**). In an idealized case where the disk solids and gas are accreted purely into the planet core and envelope, respectively, the envelope (and thus the measurable atmosphere) would have a high (supersolar) C/O ratio and a substellar C/H ratio if the planet acquired material between the H₂O and CO snow lines. While the description of this idealized case was a clear turning point in planet formation studies, dozens of studies have since built on or branched off from it to show how a multitude of other processes and factors complicate the picture and can lead to degeneracies in the interpretation of planetary atmospheric abundances (for a review, see Mollière et al. 2022).

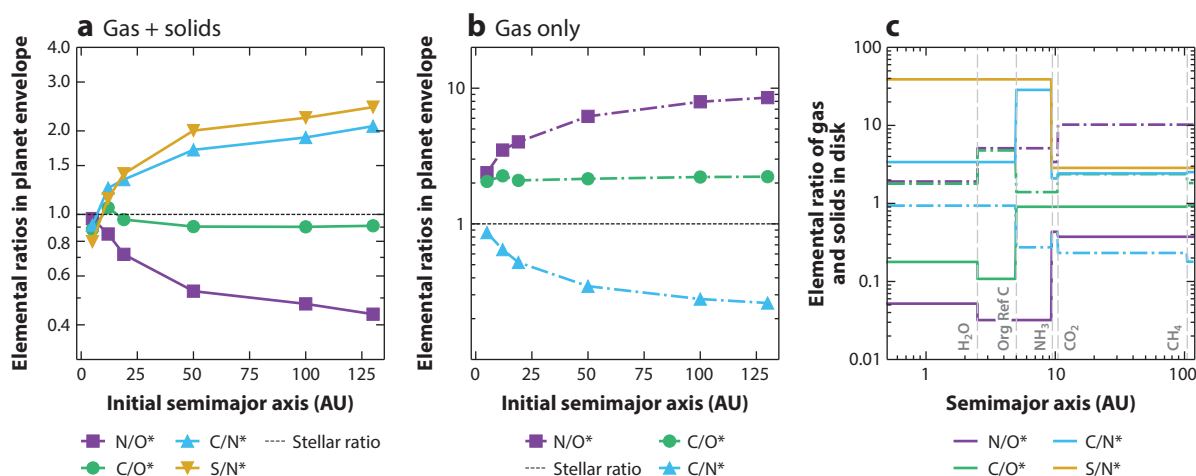


Figure 3

Elemental ratios of a giant planet's gas envelope metallicity when it is dominated by (a) planetesimal accretion or (b) gas accretion. The figure illustrates how the formation location of a giant planet can in principle be narrowed down on the basis of how its atmospheric volatile abundances compare with those of its host star. All planets are assumed to reach a final semimajor axis of 0.4 AU. Note the difference in y-axis scale between these panels. (c) Change with radial distance from the star in elemental ratios of the gas (dash-dotted lines) and solid (solid lines) components of the protoplanetary disk. In all panels, the elemental ratios are normalized to the relevant stellar elemental ratios (hence the asterisk notation). See Turrini et al. (2021) for further details (e.g., CO is missing because it does not condense due to the adopted thermal profile). Figure adapted with permission from Turrini et al. (2021); copyright 2021 AAS.

For example, a significant change to the idealized case is the addition of solids to the planet envelope via planetesimal accretion. In their “chain” of models linking planet formation, migration, thermodynamic evolution, and disk chemistry, Mordasini et al. (2016) find that the atmospheric dissolution of planetesimals is the dominant factor in determining envelope composition. This finding is motivated by the above point about giant planet metal enrichment. During formation, gas-phase heavy elements accreted with gas can contribute to the enrichment, but the authors suggest that planets with superstellar metal enrichment [Z_p/Z_* or $(N_Z/N_H)_p/(N_Z/N_H)_* > 1$] must have experienced accretion of solids. From Solar System planets, gas giant exoplanets, and theoretical planet population syntheses, this transition from gas-dominated to planetesimal-dominated accretion appears to be around $10 M_{Jup}$, although it could be slightly lower. Planetesimal accretion also seems to require migration, and total metal mass may depend on migration distance (Shibata et al. 2020). Importantly, the dominance of planetesimal accretion results in atmospheres that basically always have $C/O < 1$, no matter where in the disk the planet formed (Madhusudhan et al. 2014, Cridland et al. 2016). According to Mordasini et al., this is because the planetesimals are modeled to contain ~ 25 – 50% of refractory material (a mix of Fe, refractory C, and silicates with $MgSiO_3$ stoichiometry), with at most an equal amount of C and silicates, and the volatile component (which can be in the planetesimals as ice or in gas) is dominated by O-bearing species (e.g., H_2O always has the highest number density). Note that while a range of C/O ratios are observed in exoplanet host stars, they are almost entirely smaller than one (e.g., Teske et al. 2014, Nissen 2015, Brewer et al. 2016, Biazzo et al. 2022). These results predict that giant planet atmospheres will generally have substellar or at most stellar C/O ratios.

A key assumption made by Mordasini et al. (2016) is that C is depleted in the inner disk, which is observed in Solar System bodies (Bergin et al. 2015) and polluted white dwarfs (Section 4.2.1). If instead C depletion in the inner disk is turned off and set at a more interstellar medium-like composition, a planet formed completely inside the water ice line can indeed have a C/O ratio greater than one because the refractory (solid) materials are now more abundant in C than silicates. There could also be a small range of locations in between the C depletion radius and the ice line where planets could end up with C-rich envelopes, but this scenario is more sensitive to model assumptions. If C is depleted in (removed from) the refractory component, it may conversely be enhanced in (returned to) the gas and available for planet envelope accretion, which could also result in superstellar C/O in giant planet atmospheres (Cridland et al. 2019).

Another factor that complicates the picture is that solids in the disk are not static in size or position. Grain growth is efficient up to centimeter (“pebble”) sizes and happens on fast timescales compared with the disk lifetime (a thousand to tens of thousands of years; Dullemond & Dominik 2005, Brauer et al. 2008, Birnstiel et al. 2010). These grains are then quickly pulled inward due to gas drag, creating a pipeline of pebbles into the inner disk that are coated in H_2O , CO_2 , or CO ice, depending on their initial location (e.g., Piso et al. 2015, Booth et al. 2017, Krijt et al. 2018). As the icy pebbles drift into hotter regions of the disk, they will sublime and release vapor, which will then diffuse across a wider disk region. In pebble accretion models of planet formation, the planet core is formed first by solid accretion until the pebble insolation mass is reached, at which point gas accretion takes over, meaning that most of the accreted solids are confined to the core (Lambrechts & Johansen 2012), although they can in principle dissolve and contribute to the envelope metallicity (Ali-Dib 2017, Madhusudhan et al. 2017). The planet envelope would thus be metal-poor (subsolar C and O), and the C/O ratio would depend more on the gas composition. By including pebble drift and sublimation, supersolar C/O and C/H ratios in gas giant envelopes accreted between the CO_2 and CO ice lines (~ 10 – 40 AU) are possible without pollution by solids (Öberg & Bergin 2016, Booth et al. 2017, Schneider & Bitsch 2021a). Indeed, supersolar C/O and C/H ratios are not expected in the case of envelope pollution from solid material, so these

ratios in comparison to those of host stars may at least help distinguish planetesimal versus pebble accretion (**Figure 3**).

Reggiani et al. (2022) recently emphasized the importance of accurate and precise host star abundances in the case of hot Jupiter WASP-77A b. By updating the [C/H] and [O/H] abundances and the C/O ratio of the host star, these authors found that the planet atmosphere has substellar C/H and O/H but superstellar C/O (instead of stellar, as previously assumed; Line et al. 2021). Thus, they suggest that the planet formed beyond the H₂O ice line in the disk, instead of within the ice line and from C-depleted gas with little subsequent planetesimal accretion. More robust constraints on hot giant planet C/O ratios are still scarce, in part because of the presence of clouds and hazes (Sing et al. 2016) and in part because *Hubble Space Telescope*/WFC3 does not cover C-bearing molecular bands. Yet already there appears to be diversity in giant planet atmospheric C/O ratios [e.g., HD 149026b with C/O $\sim$ 0.80 (Bean et al. 2023) versus HD 209458b with C/O $\sim$ 0.11 (Xue et al. 2024)], and the increased capabilities of JWST will result in a larger and more precise sample of giant planet atmospheric C/O ratios, such that previously proposed first-order trends (e.g., Brewer et al. 2017, Welbanks et al. 2019) can be better tested.

However, what should be clear is that the connection between host star and planet atmospheric C/O ratios is not straightforward by any means; indeed, the utility of this ratio for distinguishing formation and migration pathways may be limited (Turrini et al. 2021). More sophisticated modeling efforts are in progress (e.g., Eistrup et al. 2018, Emsenhuber et al. 2021, Mollière et al. 2022, Pacetti et al. 2022) and can be informed by additional diagnostics beyond C and O (and Na and K). Constraining the ratio of volatiles (C, O) to higher- T_{cond} refractories (Cr, Si, Mg, Ni, Ca, Ti, Fe, Al) in at least very hot ($T_{\text{eq}} \gtrsim 2,000$ K) giant planet atmospheres is now feasible, in large part due to ground-based high-resolution studies (Gaudi et al. 2017; Hoeijmakers et al. 2018, 2020; Casasayas-Barris et al. 2019; Ehrenreich et al. 2020; Nugroho et al. 2020; Giacobbe et al. 2021). These measurements could help narrow down formation location, as a high mass fraction of refractories would be expected from formation in, or at least accretion of refractory-rich material from, the inner disk, versus a low refractory mass fraction from formation and/or enrichment in the outer disk (Lothringer et al. 2021, Schneider & Bitsch 2021b, Chachan et al. 2023). Additional volatile species are also very promising candidates for measurement in exoplanet atmospheres, although they are more difficult than C and O to measure in host stars. Both S (Tsai et al. 2023) and N (Carleo et al. 2022, Guilluy et al. 2022) species have been detected in giant planet atmospheres, and models of planet growth and migration in disks of different compositions indicate that giant planets enriched in solids should have $C/N^* > C/O^* > N/O^*$ (where the asterisks indicate normalization to stellar values), with S/N^* increasing over C/N^* the more solids are contributing to the metallicity, versus giant planets dominated by gas accretion that should have $N/O^* > C/O^* > C/N^*$ (Turrini et al. 2021, Pacetti et al. 2022) (**Figure 3**). These predictions are due to the high volatility of N (more so than C and O), and the expected early trapping of S in solids based on meteoritic data (Lodders 2010) coupled with measurements of S in the accretion-contaminated photospheres of massive young stars that host disks (Kama et al. 2019). Ultimately, it will also be important to compare host star versus planet atmosphere with planet bulk composition constraints to understand the origin of envelope enrichments (e.g., accretion, core dissolution, deep mixing). For example, Teske et al. (2019) found that the residual bulk metallicity of cool giant planets is correlated with the ratio of host star volatile to refractory elements, which could indicate that these planets acquire metals mostly from gas versus solid accretion. This hypothesis should be tested with more atmospheric observations.

Finally, this section has focused on transiting exoplanets. Relating the host star and planet atmosphere compositions for nontransiting planets is also possible via high-resolution Doppler spectroscopy, which is well suited for close-in giant planets (for a review, see Birkby 2018) and via

spectroscopy of directly imaged planets (e.g., Konopacky et al. 2013, Barman et al. 2015, Bonnefoy et al. 2016, Mollière et al. 2020, Petrus et al. 2021). For example, Mollière et al. (2022) show that including protoplanetary disk chemical evolution can lessen the migration necessary to explain the composition of the directly imaged planet HR 8799e. Thus far, these populations are limited, so perhaps it is premature to discern clear trends or comparisons with closer-in planets. However, Hoch et al. (2023) show that the C/O ratios of 10 directly imaged planet atmospheres (not relative to their host stars) span 0.7 ± 0.2 , a much narrower range than in transiting planets. Studies of directly imaged planets can be more limited by the host star, since these stars are young and/or massive, making abundance determinations very challenging (HR 8799 is especially tricky because it is a λ Boötis-type star; Wang et al. 2020).

4.2. Rocky Planets

In the context of smaller, mostly rocky planets, host star abundances have been used as a constraint in addition to mass and radius in modeling their compositions, which, depending on the mass/size of the planet, can be degenerate between Fe cores, silicate mantles, water/icy envelopes, and H/He envelopes (e.g., Valencia et al. 2007, 2013; Rogers & Seager 2010). In particular, studies have focused on using refractory elemental abundances for two reasons. First, they make up the majority of the constituents of these types of planets—95% of the bulk Earth composition is composed of Mg, Si, Fe, and O⁹ (McDonough 2003). Second, their condensation fronts ($T_{\text{cond}} \gtrsim 1,200$ K) fall into a limited region fairly close-in to the host star, where the timescale for the disk to reach equilibrium is fast. There is also good correlation between these elements as well as the ratios of Fe/Mg and Mg/Si in Solar System rocky planets (although not Mercury) and their abundances in the Sun (Unterborn et al. 2016, Wang et al. 2019). Most planet-hosting stars for which we have measured abundances of these elements range from 0.7 to 1.5 in Mg/Si molar ratios and from 0.5 to 1.1 in Fe/Mg molar ratios (e.g., Suárez-Andrés et al. 2018). As discussed in Section 1, GCE plays a role in altering these ratios over time—Fe/Mg (and Fe/Si) will generally increase as time goes on, whereas Mg/Si will decrease slightly (i.e., is lowest for the youngest stars; Nissen et al. 2017, Bedell et al. 2018).

Assuming a match between star and planet abundance ratios (discussed later in this section), the Fe/Mg ratio is indicative of the relative size of a planet's core, and the Mg/Si ratio indicates mantle mineralogy (i.e., the distribution of Mg and Si between the minerals pyroxene, olivine, bridgmanite, and ferropericlasite) (for a schematic, see **Figure 4**). Whether the Mg/Si ratio is high (low) due to high (low) Mg or low (high) Si also has implications for the oxidation state and, thus, the core size of the planet (how much Fe is available for the core; Unterborn & Panero 2017). Note that the exact distribution of Mg, Si, Fe, and O within a given planet is uncertain, as the last three elements in this list can be incorporated into the metallic core, although Fe dominates preferentially; their distribution will depend on the processes involved in the planet's differentiation (for more details, see Section 4.2.1, below). Light alloys in the core and even the process of differentiation probably have small effects on major observables like exoplanet radius (Grasset et al. 2009, Plotnykov & Valencia 2020), but they can have significant implications for processes like magnetic field generation, plate tectonics, and outgassing (e.g., Unterborn & Panero 2017, Shahar et al. 2019).

Multiple investigations have shown compelling evidence that we should indeed be able to rely on host star Fe, Mg, and Si abundance ratios as proxies for those in forming planets, both by correlating them with the output of models of equilibrium condensation from the stellar nebula

⁹O can be in refractory form, condensing in silicates and oxides, or in volatile form, condensing as ice. This means that the assumption of stellar and planetary similarity cannot hold for O.

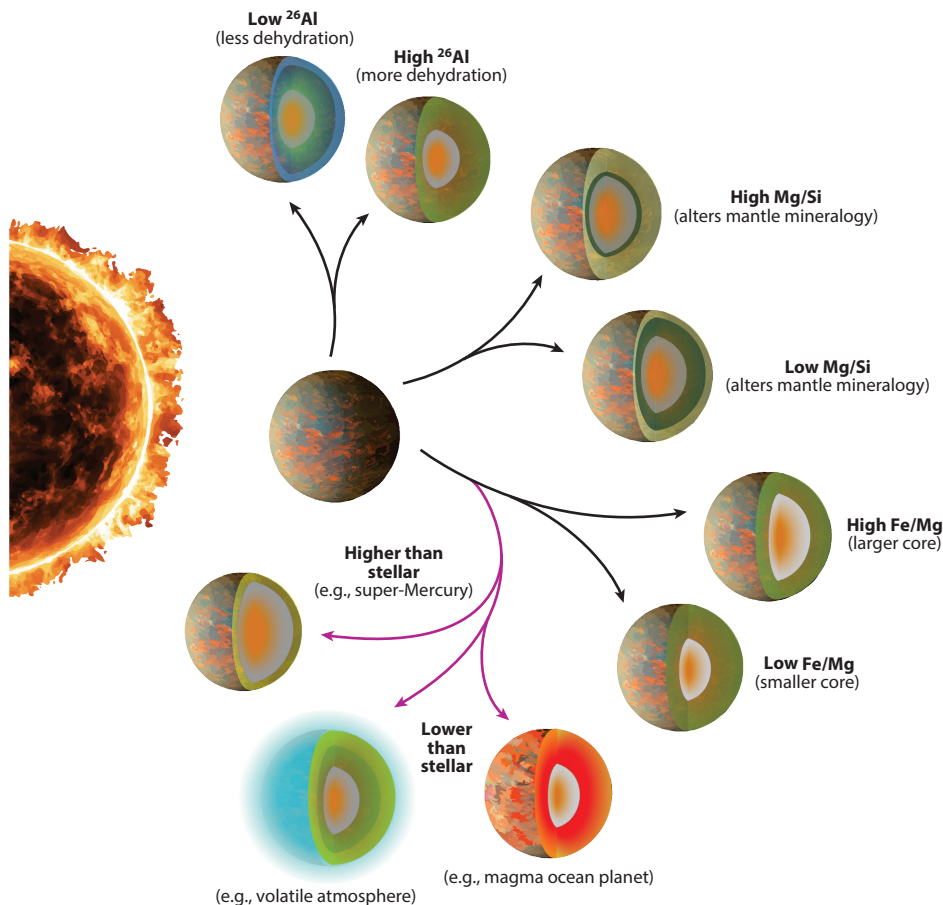


Figure 4

A rough schematic of some of the effects that different elemental ratios might have on rocky planet compositions/structures, including water content, mantle mineralogy, and core mass fraction. The figure illustrates that within the bounds set by stellar abundances, a diversity of planets are expected. There is also evidence to suggest at least some instances in which stellar and inferred planet compositions do not match (*purple arrows*); whether this is often the case is an open question. The mantle compositions derived from different Mg/Si ratios are from Spaargaren et al. (2020). Figure created by Dr. Francesca Miozzi, with inspiration from Bond et al. (2010).

(e.g., Thiabaud et al. 2015) and by showing how well-constrained properties of Solar System planets like mantle composition and core radius are better matched with interior structure models that use the solar abundances as priors (Dorn et al. 2015). Therefore, numerous studies have applied this assumption to infer the likely compositions of terrestrial planets (Brugger et al. 2017, Hinkel & Unterborn 2018, Wang et al. 2019, Spaargaren et al. 2023),¹⁰ although whether it would be

¹⁰Wang et al. (2019) and Spaargaren et al. (2023) discuss the importance of using devolatilized abundances that have been corrected according to the trend observed in Solar System rocky bodies relative to the Sun. For example, Mg, Si, and Fe are all depleted in rocky bodies by $\sim 10\text{--}20\%$. However, this correction assumes that whatever devolatilization process happened in the Solar System also happened in other planetary systems, which is uncertain.

possible to actually distinguish mineralogies that are significantly different is not clear (Hinkel & Unterborn 2018).

However, even these studies pointed out limitations/cautions of the so-called star $\equiv$ planet assumption. Close-in to the star (within the water ice line), different refractory-bearing species may have nonstellar values as a result of different thermodynamic properties and thus condensation fronts; if the short-period planets that we observe today do not migrate far and instead form via a more in situ mechanism, this could still cause variation in their refractory ratios away from stellar values. Additionally, with regard to small planets that could contain significant volatile components—icy layers, oceans, atmospheres—the limiting factor in determining more than just the mantle composition (e.g., water mass fraction, envelope metallicity) remains uncertainties in planet mass and radius (e.g., Dorn et al. 2017).

Furthermore, recent research has called the star $\equiv$ planet assumption into question by focusing on planets that are very likely to be rocky, where the degeneracies in composition calculated from mass and radius alone are minimized, and by comparing the core mass fraction (or Fe/Si and Fe/Mg) derived solely from the average planet density (and an interior structure model) with those predicted by the stellar abundances (Plotnykov & Valencia 2020, Schulze et al. 2021). In most direct comparisons, the host star and orbiting planet are consistent but with wide errors, so it is hard to be confident in a match. Plotnykov & Valencia (2020) also performed a broader population-wide comparison between all small exoplanets with well-constrained mass and radius data (errors below 25%) and the core mass fraction as well as Fe/Mg and Fe/Si ratios derived from stellar abundances of all planet-hosting stars in the Hypatia catalog (Hinkel et al. 2014), a database of stellar abundances sourced from the literature. They found that the planet-derived chemical ratios and core mass fractions span a wider range than the star-derived values and do not peak at the same values. In two of the systems discussed by Schulze et al. (2021), the planet density– and stellar abundance–derived core mass fractions are inconsistent at the 2σ level. One may be a so-called super-Mercury, although how these planets may form is unclear; collisions do not seem to explain them (Scora et al. 2020). The other system has a significantly lower density than expected, perhaps as a result of volatile outer layers, or depletion in Fe or more of it incorporated into the mantle versus the core. Both Schulze et al. (2021) and Plotnykov & Valencia (2020) suggest that the cases where there is clear disagreement between the planet- and star-predicted properties could be particularly constraining for small planet formation models (for a few examples, see **Figure 4**).

Admittedly, the findings above are somewhat puzzling in the context of the compelling correlation shown by Adibekyan et al. (2021): The mass fraction of iron present in the protoplanetary disk, predicted from host star abundances, is positively correlated with the bulk density for $R_p < 1.6 R_\oplus$ short-period exoplanets, and this correlation is not explained by atmospheric escape. [Note, however, that Plotnykov & Valencia (2020), Schulze et al. (2021), and Adibekyan et al. (2021) use different approaches for inference/modeling of the star/disk versus planet compositions.] Overall, to more robustly assess the star $\equiv$ planet assumption, more research is needed to increase the sample size of planets with both precise masses and radii and precise host star abundances. Another new area of emerging research is the study of how small planet interiors interact with, and thus alter the composition of, their atmospheres (Kite et al. 2016, 2020; Guzmán-Mesa et al. 2022), which may result in testable predictions between host star and small planet atmospheric compositions, similar to those we have for giant planets (Section 4.1).

Aside from the proportions of major refractory species, another factor controlling small planet composition is radiogenic elements that contribute to internal heating. Early on in planetary system formation, decay heat from the short-lived radionuclide ^{26}Al can contribute to silicate melting and degassing of primordial water in the planetesimals that accrete to form rocky planets, creating a gradient of water mass fractions that, at the extremes, could result in a difference

in radius of up to 10% (Lichtenberg et al. 2019; note that this model assumes that small planet formation is dominated by planetesimal accretion). I discuss the possible constraints on ^{26}Al from stars more in Section 4.2.1 but note here that Lichtenberg et al. (2016) used simulations of young star clusters to suggest that 10–30% of planetary systems can be enriched in ^{26}Al at values similar to what is inferred for the Solar System. Later on in the life of a planet, the decay of long-lived radioactive isotopes of K, Th, and U can also dictate its internal energy budget and thus processes like mantle convection and volatile recycling, and there appears to be enough variation in Th (a refractory element produced by the *r*-process, likely in neutron star mergers; Kasen et al. 2017) in comparison to Si (as a mantle proxy) in solar twin stars to imply significant differences in planetary thermal evolution (Unterborn et al. 2015).

4.2.1. Polluted white dwarfs. This review has focused almost exclusively on the relationship between the compositions of main-sequence stars and exoplanets, but it would be remiss to leave out the important information about planet composition derived from polluted white dwarfs. These remnants of solar-type stars have extremely high surface gravities that cause elements heavier than H and He to sink rapidly (days to years for H-dominated white dwarfs) out of their observable photospheres, and yet 25–50% of white dwarf spectra show UV and optical absorption lines of metals like O, Mg, Si, Ca, and Fe—these elements must come from an external source (Zuckerman et al. 2010, Barstow et al. 2014, Koester et al. 2014). Some but not all polluted white dwarfs also show IR excess indicating a circumstellar disk of debris (see Farihi et al. 2009, Barber et al. 2012, Rocchetto et al. 2015, and references therein). After several decades, the consensus picture explaining the pollution signature is that asteroids, comets, and/or planetary bodies that orbited the main-sequence star are dynamically scattered close enough in to the white dwarf to be tidally disrupted, sometimes settling into a disk of dust and gas, and accreted (Debes & Sigurdsson 2002, Jura 2003). This means that polluted white dwarfs provide unique constraints on the compositions of extrasolar planetesimals that rely on fewer assumptions about how host star and planet material are related, making them a powerful tool for understanding the diversity of exoplanet compositions. For excellent reviews of polluted white dwarfs, their circumstellar disks, and the resulting implications for planetary systems, see Jura & Young (2014), Farihi (2016), and Veras (2021). This section provides a brief synopsis and a few new perspectives from recent research.

First, direct evidence in support of the picture described above has been detected in multiple systems: Several examples of rocky, disintegrating debris transiting in front of white dwarf stars have been found (Vanderburg et al. 2015; Vanderbosch et al. 2020, 2021). Most polluted white dwarfs, including those with detected orbiting debris, are consistent with having accreted material resembling the bulk Earth in that O, Mg, Si, and Fe dominate and there is little evidence for water or C-bearing species (Jura & Xu 2012; Jura & Young 2014; Xu et al. 2014, 2019). Water is abundant in planet-forming regions (e.g., van Dishoeck et al. 2021) and in principle could survive in moderately sized planetesimals even during the star's red giant branch stage (Jura & Xu 2010, Malamud & Perets 2016), so the dry nature of most of the debris polluting white dwarf atmospheres is somewhat surprising but helps reveal the origin and processing of the accreted material (discussed later in this section).

There have been a few interesting exceptions to the inferred dry, Earth-like compositions, including several likely water-rich bodies (Farihi et al. 2011, 2013; Raddi et al. 2015; Hoskin et al. 2020), one of which appears to be an N-rich Kuiper Belt Object analog (Xu et al. 2017); a possible evaporating icy giant planet (Gänsicke et al. 2019), likely thrown close-in to the star by dynamical interactions with other planets, the nature of which was bolstered by the detection of an actual transiting giant planet candidate around a white dwarf star (Vanderburg et al. 2020); and a possible icy exomoon formed in the rings around a giant planet, as evidenced by a significant excess of Be

that is most likely produced by a high proton fluence in the radiation belts around the giant planet (Doyle et al. 2021, Klein et al. 2021). Notably, the debris around the majority of white dwarfs does not appear to be C-rich and generally has C/O ratios well below solar (Wilson et al. 2016).

With increasing numbers of metals detected in polluted white dwarfs, distinctions within so-called Earth-like accreted material have become possible. These more detailed observations have been used to suggest differentiation occurred in extrasolar rocky bodies—white dwarfs show polluted material that is core-like (Fe- and/or Ni-rich) versus mantle-like or crust-like (Fe-poor and Ca-, Al-, and/or Ti-rich) (Jura et al. 2013, Melis & Dufour 2017, Harrison et al. 2018, 2021, Hollands et al. 2018, 2021; but see Brouwers et al. 2023a,b for caution about how asymmetric accretion can bias the interpretation of the polluted material bulk composition). For the planetesimals polluting white dwarfs to form cores and mantles, they must have experienced substantial heating. In planetesimals larger than $\sim 1,500$ km, their gravitational potential energy can provide the requisite energy for large-scale melting, but smaller bodies need a different source. In the Solar System, this source is the decay of the short-lived radionuclide ^{26}Al , thought to be injected into the solar nebula by a nearby supernova (Davis et al. 2014).

The half-life of ^{26}Al is 0.717 Myr, and its effectiveness as a heating source decreases after one to two half-lives. Therefore, if planetesimals accreted by white dwarfs are differentiated, and the heating that causes this differentiation comes from ^{26}Al , then the planetesimals must have formed relatively early, within the first 1–2 Myr after molecular cloud collapse. The presence of differentiated material in polluted white dwarfs therefore helps corroborate the picture from observations of substructure in disks around very young stars that planet formation began early [perhaps in the class 0/I stage (Harsono et al. 2018, Cridland et al. 2022, Pinte et al. 2023); but see also recent results from the ALMA eDisk program (Ohashi et al. 2023)] in at least some systems (Bonsor et al. 2023). Furthermore, if a large fraction of polluted white dwarfs show evidence of accreting differentiated material, it would indicate that either most of the material is from Pluto-sized or larger bodies (unlikely; Curry et al. 2022) or that most systems have short-lived radionuclide enrichment (implying that the Solar System’s initial abundance of ^{26}Al was fairly normal; Jura et al. 2013, Jura & Young 2014, Curry et al. 2022).¹¹ By studying the remnants of planets around stars at the ends of their lives, we thus gain insight into the first stages of planet formation around our Sun and other stars. Heating early on in the formation of planetary systems has significant implications for how volatiles are distributed, for instance, whether small planets are water depleted, as in the Solar System, or water-rich “ocean worlds” (Lichtenberg et al. 2019, Adams 2021, Lichtenberg & Krijt 2021).

The variety of polluted white dwarf abundances has also been framed in the context of O fugacity (f_{O_2}), or the effective partial pressure of O gas that would be in thermodynamic equilibrium with the material of interest, of the accreted debris. Oxygen fugacity controls how Fe is divided between metal and rocky minerals and, thus, the core fraction (and whether it contains light elements) and geochemistry of the mantle and crust. These factors then influence the presence and composition of a planet’s atmosphere, the strength of a planet’s magnetic field, and the likelihood of plate tectonics (see, e.g., Foley & Driscoll 2016, Shahar et al. 2019, and references therein). The majority of the Solar System rocky bodies (Mars, Earth, Vesta, and most chondrite species, but not Mercury and enstatite chondrites) formed in oxidizing conditions relative to H-rich solar composition gas, although Earth is on the less oxidized (more reduced) end of this distribution. There are still challenges in confidently detecting low- f_{O_2} material in polluted white dwarfs, but so far the composition of most material has been consistent with the range of f_{O_2} Solar System values (fairly

¹¹And/or it indicates that enrichment of systems by ^{60}Fe is also common.

oxidized; Doyle et al. 2019, 2020, 2023). Interestingly, the oxidized nature of polluted white dwarf material may point toward an important role for H_2O after all, as its presence could increase f_{O_2} in the dust that forms planetesimals, and/or aqueous alteration in undifferentiated parent bodies could transform Fe metal into Fe oxides, further increasing f_{O_2} (Krot et al. 2000, Monteux et al. 2018).¹² Note that the extent and timing of ^{26}Al heating also come into play here—some heating helps create hydrated minerals, but as noted above it can also totally dehydrate planetesimals (Monteux et al. 2018, Lichtenberg et al. 2019).

Several studies have compared the abundances or rock mineralogies implied from polluted white dwarf measurements with those implied from nearby main-sequence stars in order to test whether the compositional spreads are similar (Harrison et al. 2021, Putirka & Xu 2021, Trierweiler et al. 2023). Given the uncertainties in both the white dwarf and stellar abundances (Figure 5), it is difficult to say whether there is actually more variation in one group or another, or whether some polluted white dwarf compositions imply exotic compositions in comparison to nearby stars. Similarly, a direct comparison between the refractory abundances in the polluted white dwarf WD 1425+540 and its wide (projected separation $\sim 2,240$ AU) binary K dwarf

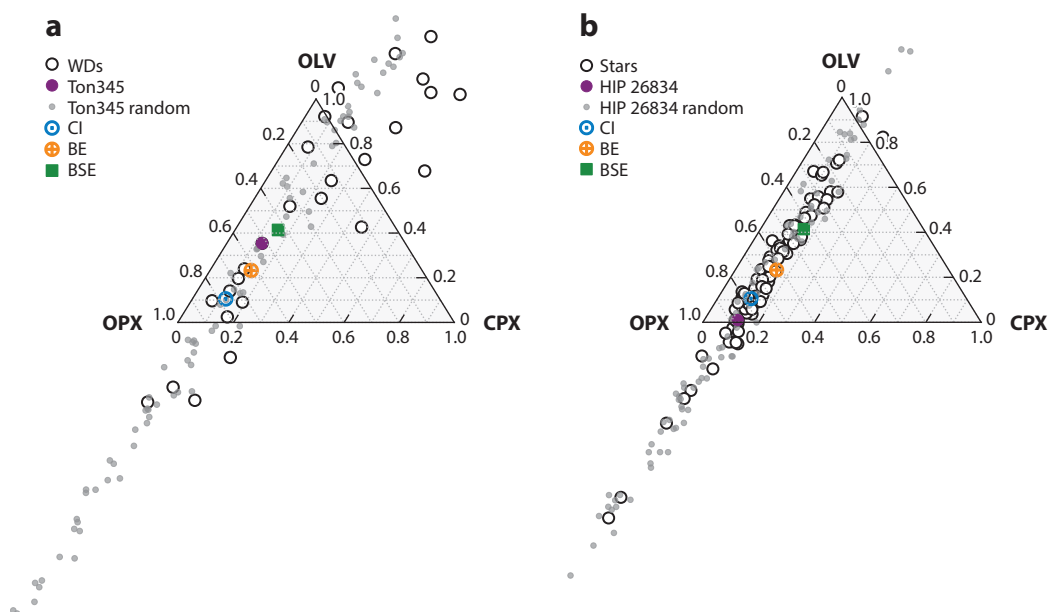


Figure 5

Ternary diagrams of relative volume fractions of normative mineralogies $\text{OLV} = \text{Mg}_2\text{SiO}_4$, $\text{OPX} = \text{Mg}_2\text{Si}_2\text{O}_6$, and $\text{CPX} = \text{CaMgSi}_2\text{O}_6$ calculated from abundances of (a) polluted WDs from the literature (*white points*) and (b) randomly selected stars from the Hypatia catalog (*white points*). Gray points represent the spread from 100 random draws within the errors of Mg, Si, and Ca for (a) WD Ton345 and (b) HIP 26834, both of which have abundances that are good matches to CI chondrites. The figure illustrates that the large uncertainties on existing WD abundances (particularly Si and Mg) make it challenging or impossible to place robust constraints on the implied rock mineralogy. The volumetric fractions results are not necessarily physical, resulting in negative amounts of OLV, OPX, or CPX in this projection. Also shown for reference are CI chondrites, BE, and BSE. Abbreviations: BE, bulk Earth; BSE, bulk silicate Earth; CPX, clinopyroxene; OLV, olivine; OPX, orthopyroxene; WD, white dwarf. Figure adapted from Trierweiler et al. (2023) (CC BY 4.0). Data provided by the author, including a different random draw for stars and WDs.

¹²Part of the challenge of finding more reduced materials among polluted white dwarfs could be that more reduced planetesimals initially formed closer to the star, so their daughter products were more likely destroyed during the stellar evolution process (Rubie et al. 2015).

companion G200–40—serving as a proxy for the progenitor of WD 1425+540—finds that the abundances overlap but, given uncertainties, cannot firmly conclude that the planetary material accreted by the white dwarf has the same composition as that of the K dwarf star (Bonsor et al. 2021).

Overall, then, from polluted white dwarfs we can infer that rocky exoplanets are probably not that different in composition from rocky Solar System material, although they span the range measured within the terrestrial planets, minor bodies, and chondrites that are all derived from the same initial protoplanetary material but experienced different levels or types of processing during formation (Swan et al. 2019, Trierweiler et al. 2023).

5. FUTURE PROSPECTS

The next phase of studying the connection between stellar and planetary compositions will be greatly influenced by the influx of new observations from *Gaia*, JWST, and large ground-based spectroscopic surveys like SDSS-V (Kollmeier et al. 2019), 4MOST (de Jong et al. 2019), and WEAVE (Jin et al. 2024). Some significant questions that can be (better) addressed in the coming years include the following:

1. How do the abundances in stars that host Solar System analogs—small inner planets with large outer planets—compare with the solar abundance trend? *Gaia* DR4 and DR5 are expected to provide astrometric detections of thousands of new long-period (0.5–5 AU separation) giant planet candidates (Sozzetti & de Bruijne 2018) that are challenging to find via radial velocities alone. Some of these will be in systems with known inner planets, and others around bright stars can motivate increased radial velocity monitoring for discovering new inner small planets.
2. What is the lower metallicity limit for planet formation? Occurrence rate studies with TESS have already begun to address this question for the shortest-period giant planets (Boley et al. 2021), and efforts to perform more complete occurrence rate studies are in progress (even though TESS was not designed for this purpose!). The recent commissioning of new extreme-precision radial velocity spectrographs on large telescopes (e.g., VLT/ESPRESSO, Keck/KPF) also opens up the possibility of following up or detecting planets around farther-away and/or more metal-poor stars. An intriguing synergy between exoplanet and dark matter hunters is the search for planets around stars identified by *Gaia* to be likely members of stellar streams, tidally disrupted globular clusters, or dwarf galaxies, which could have provided significantly different chemical environments for planet formation (e.g., Yoshida et al. 2022).
3. Do the trends we see in FGK host star abundances continue into the smaller M dwarf regime? How will our perspective on the diversity of small planet compositions change when we can measure detailed abundances in more of their host stars (as many known small planets orbit M dwarfs)? Some investigations into these questions have already occurred (e.g., Souto et al. 2018, Rodríguez Martínez et al. 2023), though they were limited in sample size and/or which elements are included. The Milky Way Mapper survey component of SDSS-V is planning to obtain $R \sim 22,500$ H -band spectra with APOGEE-N and -S, well suited for measuring M dwarf abundances, of a few hundred thousand short-cadence and planet-host TESS targets, as well as TESS Objects of Interest without detected planets as a control sample.
4. Where and when in protoplanetary disks do giant planets form? Is there a dominant formation pathway, or multiple pathways? Key ingredients here are a larger total number and range of species covered in exoplanet atmospheric observations with JWST and/or

ground-based high-contrast direct imaging spectroscopy, as well as more measurements of refractory species in hot planet atmospheres via space-based UV or ground-based high-resolution optical spectroscopy. Comparing the planet abundances with those in their host stars, versus assuming a solar host star composition, will be important as we build up a larger and more precise sample of planet atmospheric compositions.

5. Just how similar are rocky planet refractory compositions to their host stars? The larger sample of small planet host star abundances mentioned above will help here, but another relevant perspective may come from “hot rock” or “lava world” planets, orbiting so close to their host stars that they should not have gaseous envelopes and, thus, are amenable to surface-composition constraints from IR emission and phase-curve observations (e.g., Kreidberg et al. 2019). Several such programs were already approved for JWST cycles 1–3, including NIR spectroscopic phase-curve observations of an ultrashort-period planet around an old, metal-poor, α -enhanced thick-disk star.
6. What is the balance between oxidized and reduced compositions in extrasolar rocky bodies, and what does it imply for the habitability-related characteristics (e.g., core size, volatile budget, mantle viscosity) of existing exoplanets? Polluted white dwarf abundances can provide constraints on the geochemical nature of exoplanets, and thus far, most are consistent with compositions of a similar oxidation state as most primitive Solar System material, but the sample with a suite of abundances measured is still limited. Additionally, high-confidence detection of the accretion reduced bodies is affected by more-uncertain factors (e.g., the disk lifetime and timing of accretion), and the current sample could be biased against such detections (Doyle et al. 2020); a large sample of polluted white dwarfs, particularly those with debris disks, would be helpful. Luckily, *Gaia* has already provided many thousands of high-confidence white dwarf candidates (Gentile Fusillo et al. 2021) ripe for spectroscopic follow-up to search for signatures of pollution (Hollands et al. 2021, Doyle et al. 2023). This question may also be addressed by the exoplanet surface-composition constraints mentioned above.

These and other future projects exploring the star–planet compositional connection will benefit greatly from continued coordination and collaboration among the stellar astrophysics, exoplanet, and planetary (including Earth) science communities.

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